

Chapter 28

Protists

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

Neil Campbell and Jane Reece

Lectures by Chris Romero, updated by Erin Barley with contributions from Joan Sharp

Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings

Overview: Living Small

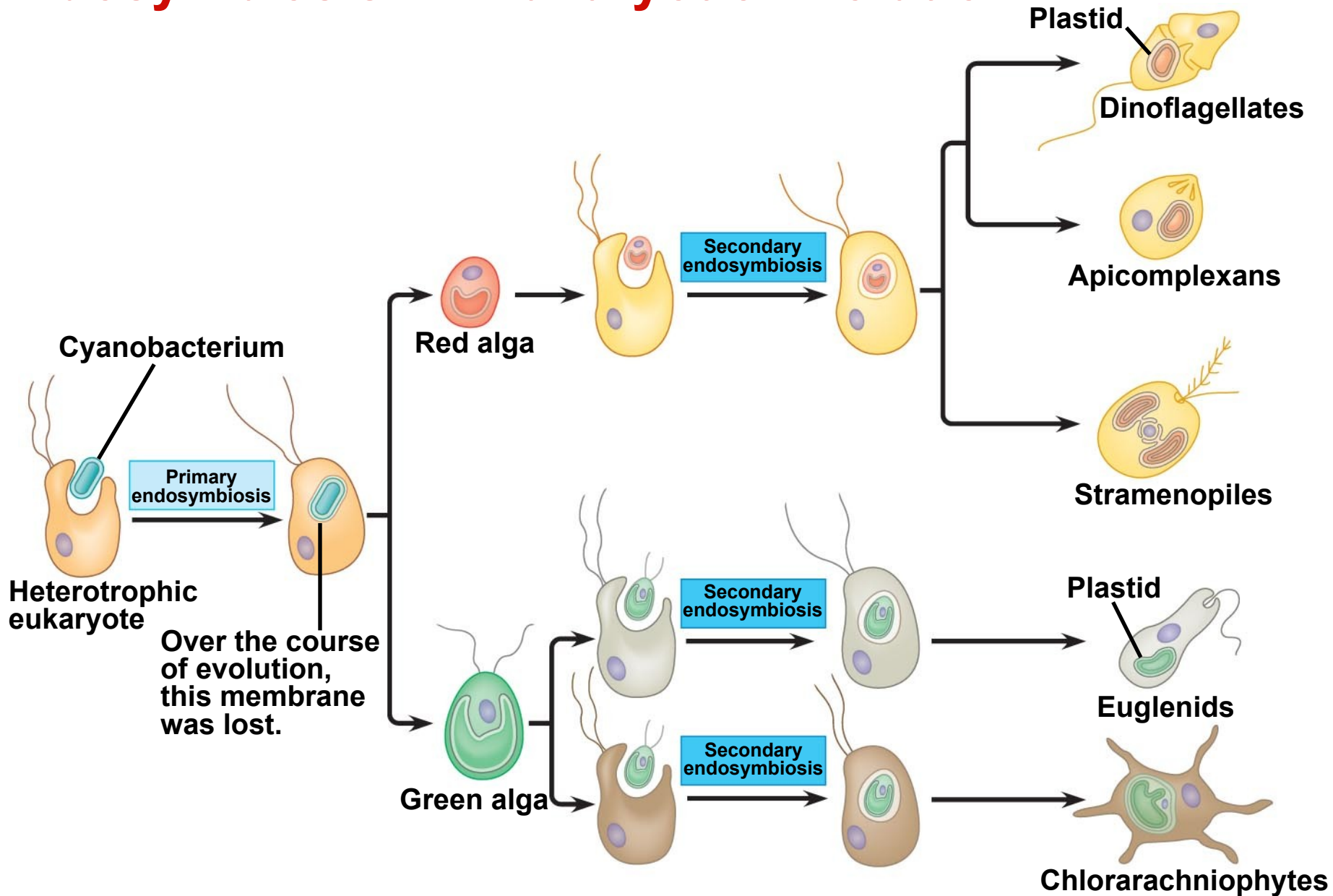
- Even a low-power microscope can reveal a great variety of organisms in a drop of pond water.
- ***Protist*** is the informal name of the *kingdom* of *mostly unicellular eukaryotes*, but there are some colonial and multicellular species.
- Protists constitute a paraphyletic group, and *Protista* is *no longer valid* as a *kingdom*.
- *Protists exhibit more structural and functional diversity than any other group of eukaryotes.*

-
- *Protists* can *reproduce asexually* or *sexually*, or by the sexual processes of meiosis and syngamy.
 - *Protists*, the *most nutritionally diverse* of all eukaryotes, include:
 - *Photoautotrophs* - contain chloroplasts.
 - *Heterotrophs* - absorb organic molecules or ingest larger food particles.
 - *Mixotrophs* - combine photosynthesis and heterotrophic nutrition.

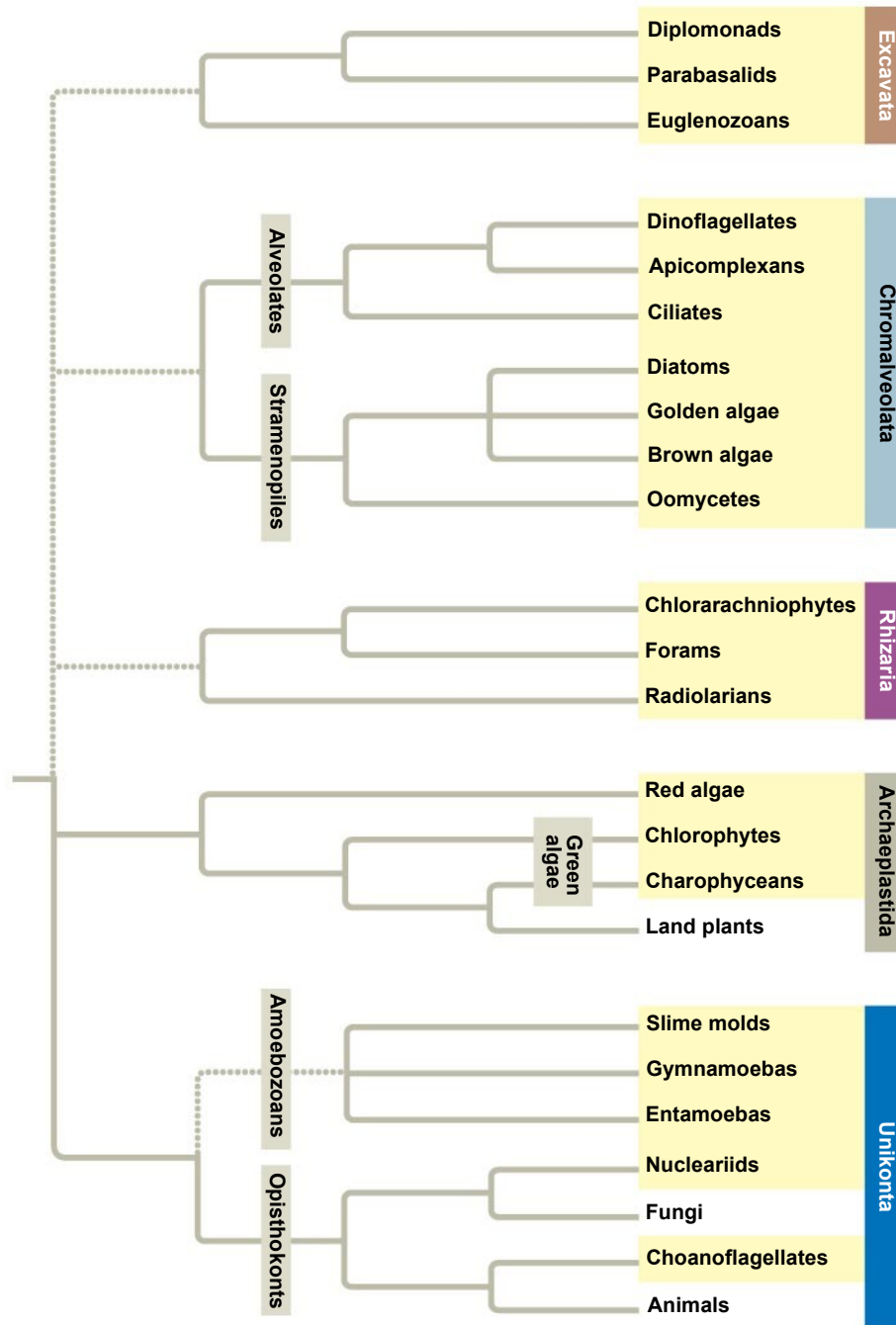
Endosymbiosis in Eukaryotic Evolution

- There is considerable evidence that much protist diversity has its origins in *endosymbiosis*.
- *Mitochondria* evolved by endosymbiosis of an aerobic prokaryote.
- *Plastids* evolved by endosymbiosis of a *photosynthetic cyanobacterium*.

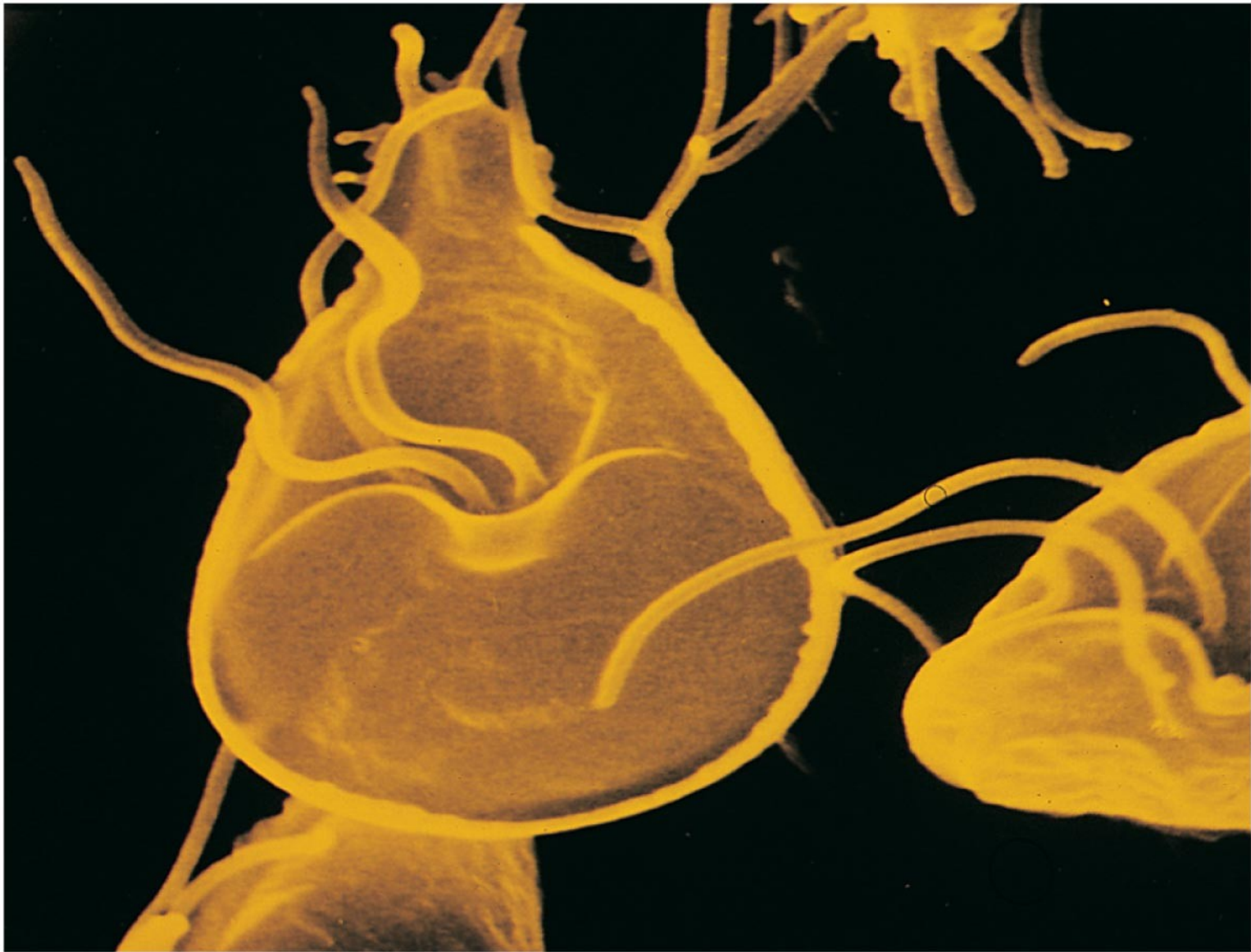
Endosymbiosis --> Eukaryotic Evolution



Protist Diversity

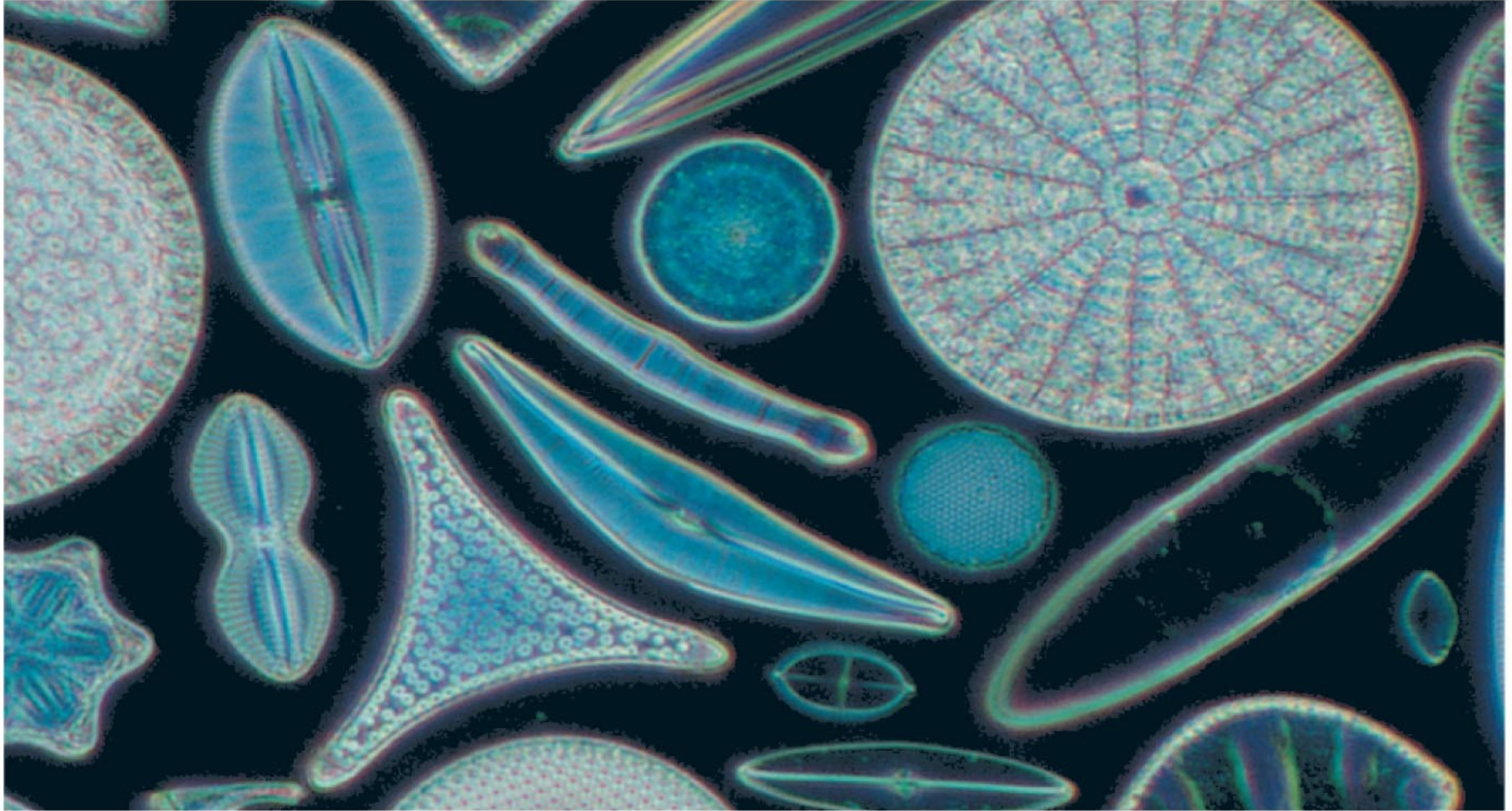


Protist Diversity - Diplomonads intestinal parasites



5 μm

Diatoms - unicellular algae / cell walls are glass-like and made of silica

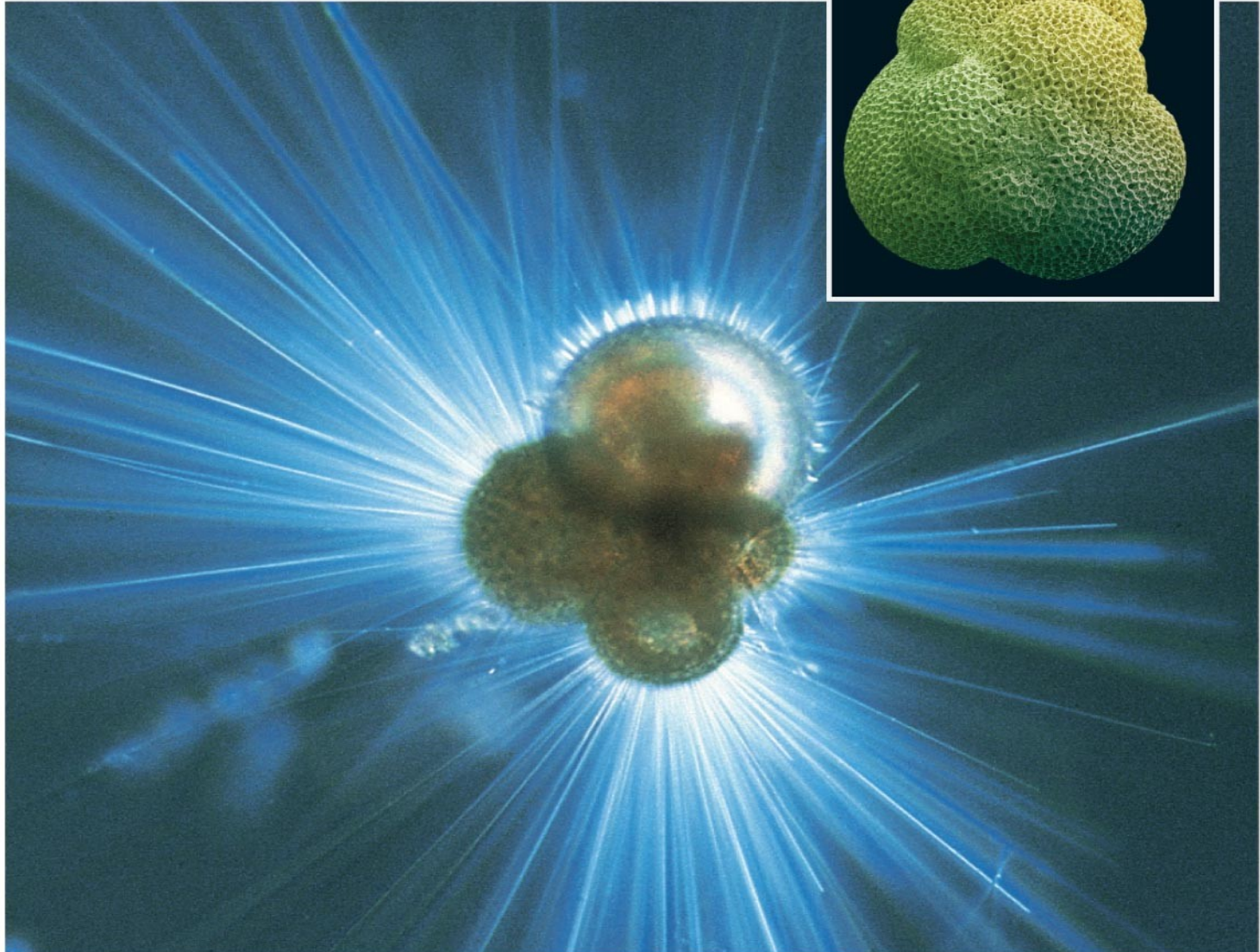


50 μm

Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings.

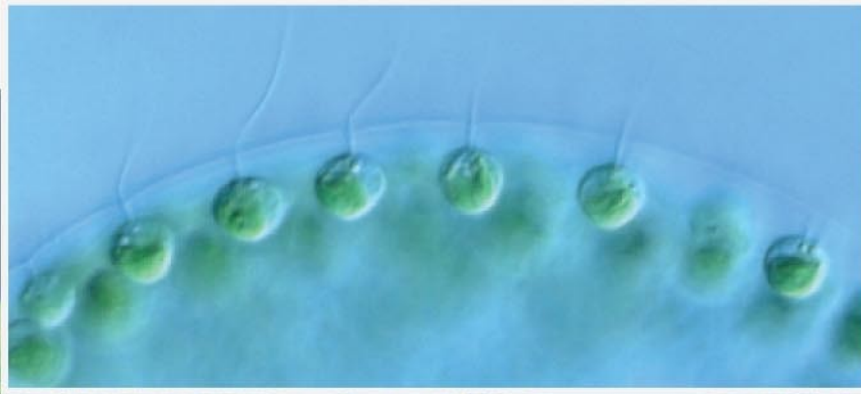
Forams - have **porous shells** called **tests**

20 μm

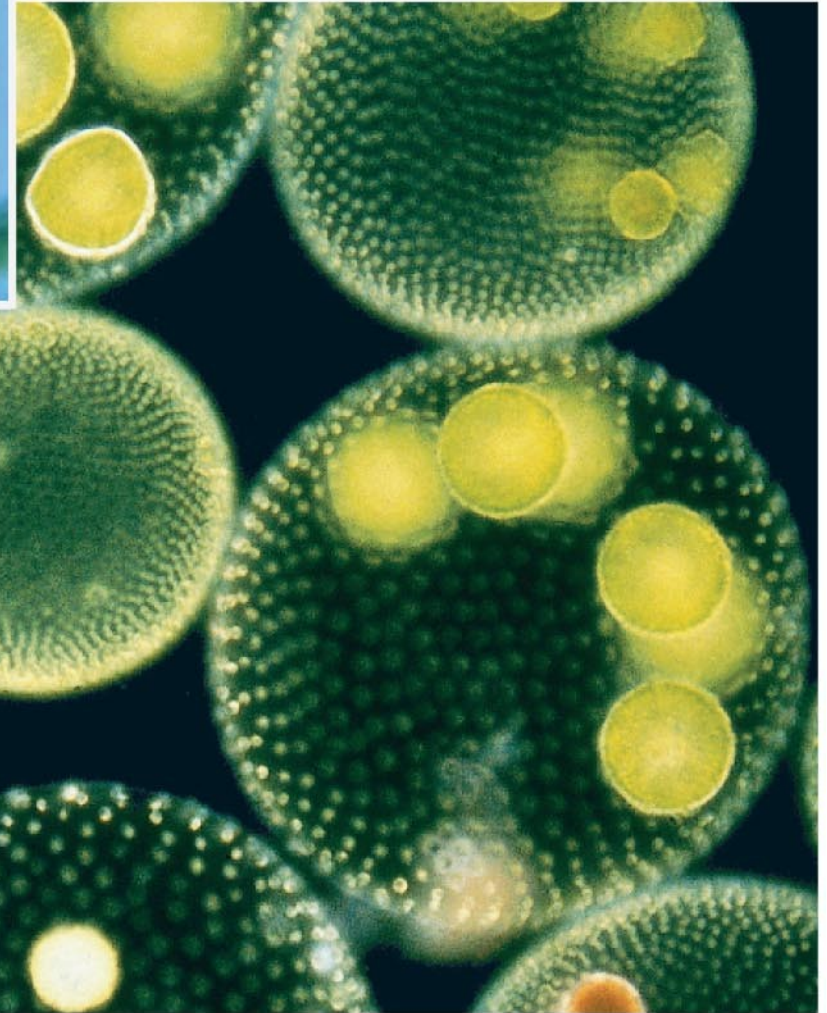


Volvox - colonial freshwater chlorophytes

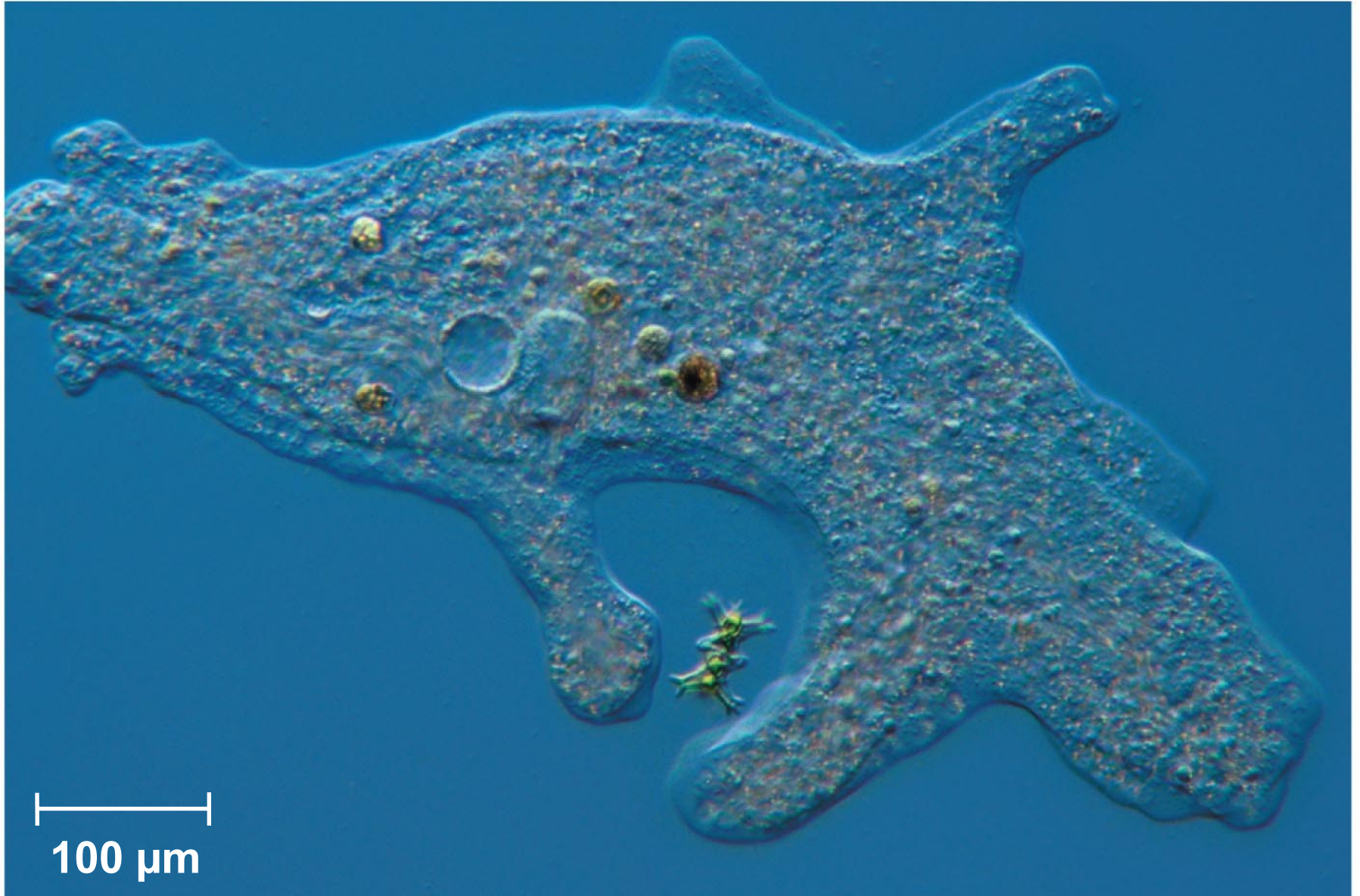
20 μm



50 μm



Amoeba - have **pseudopods** for ingestion and motility



Trichomonas vaginalis

Parabasalids that commonly inhabit the **female vagina**

Flagella



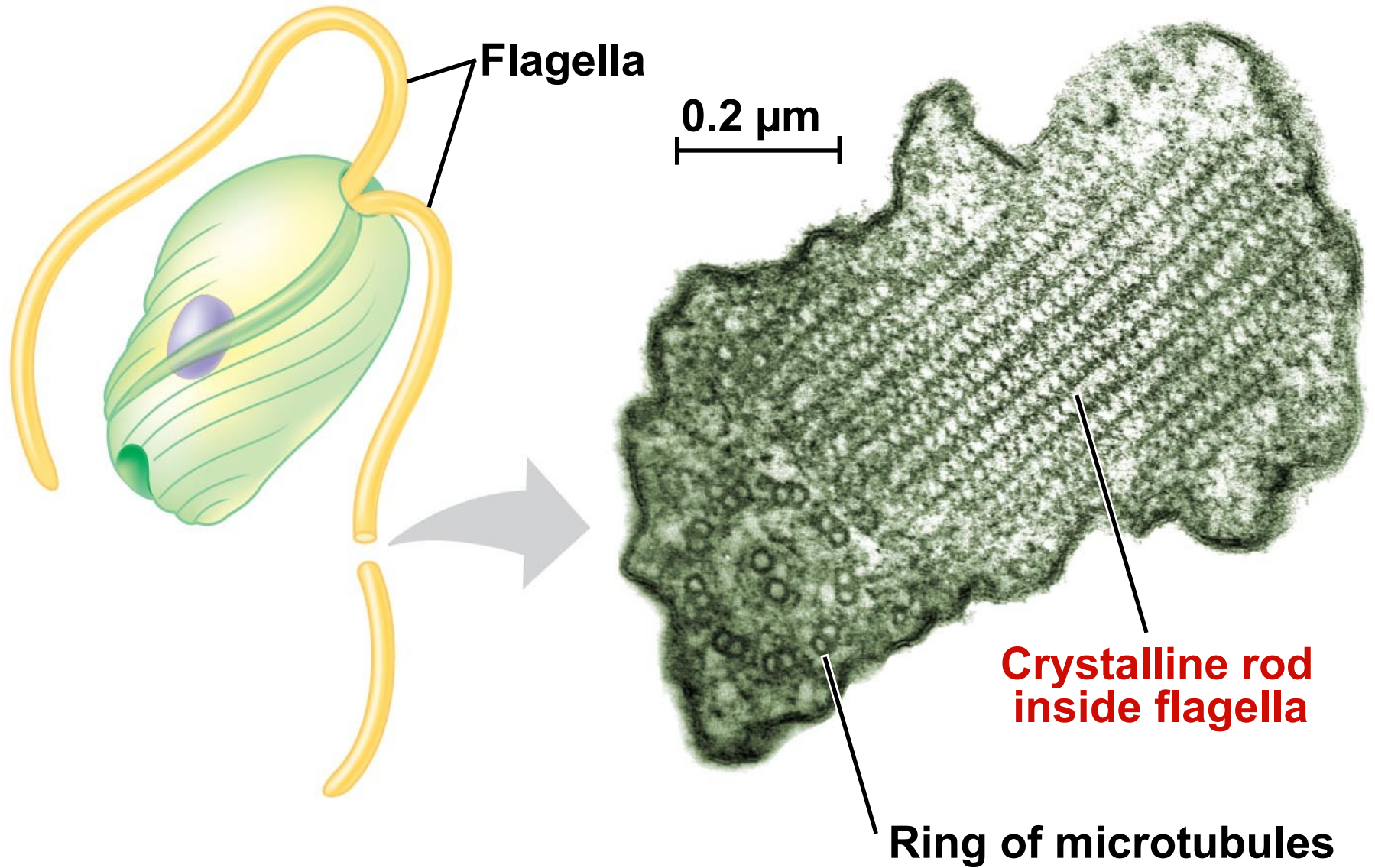
Undulating membrane

5 μ m

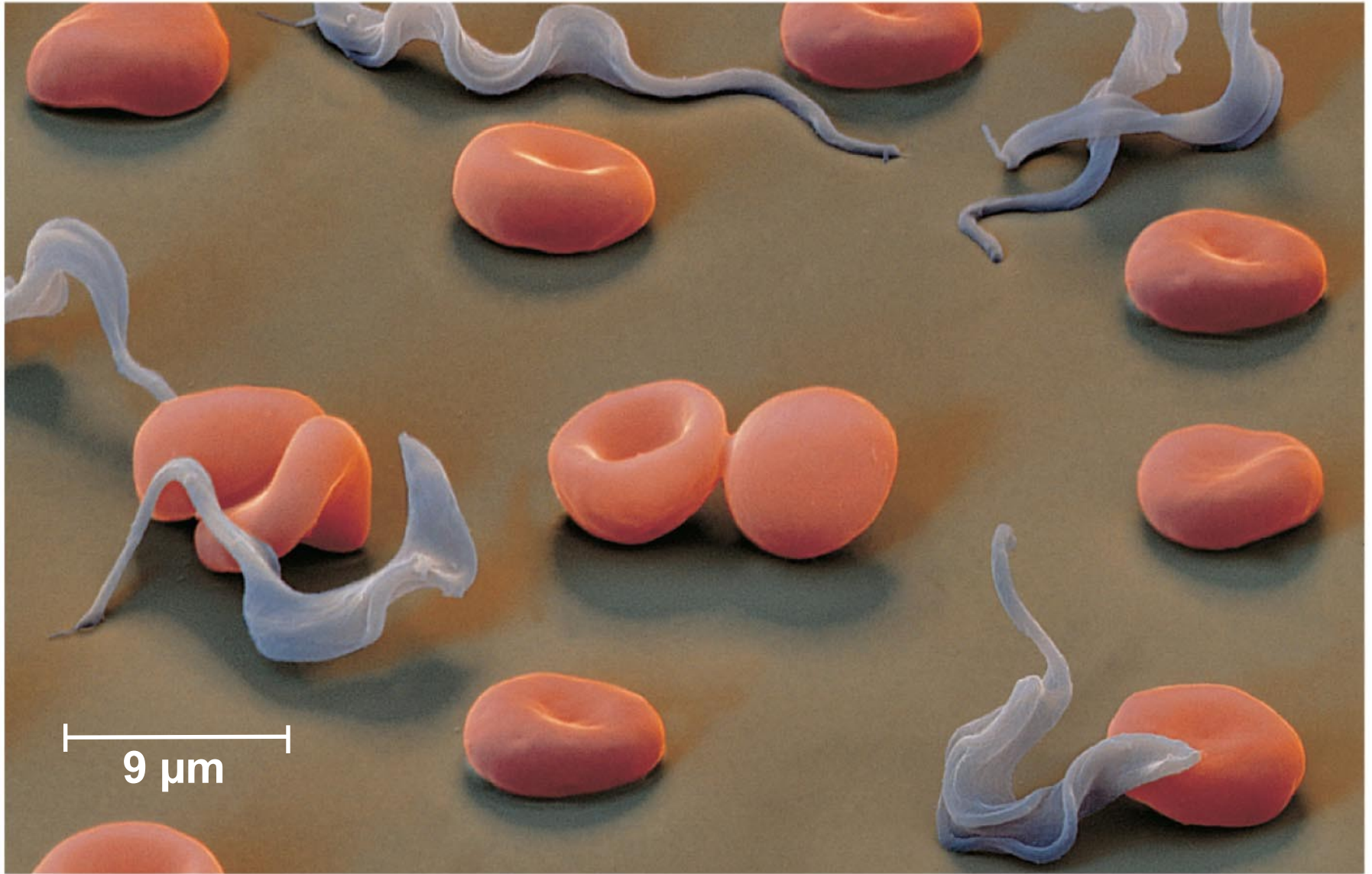
Euglenozoans

- ***Euglenozoa*** is a diverse clade that includes predatory heterotrophs, photosynthetic autotrophs, and pathogenic parasites.
- The main feature distinguishing them as a clade is a spiral or crystalline rod of unknown function inside their flagella.
- This clade includes the kinetoplastids and euglenids.

Euglenozoans

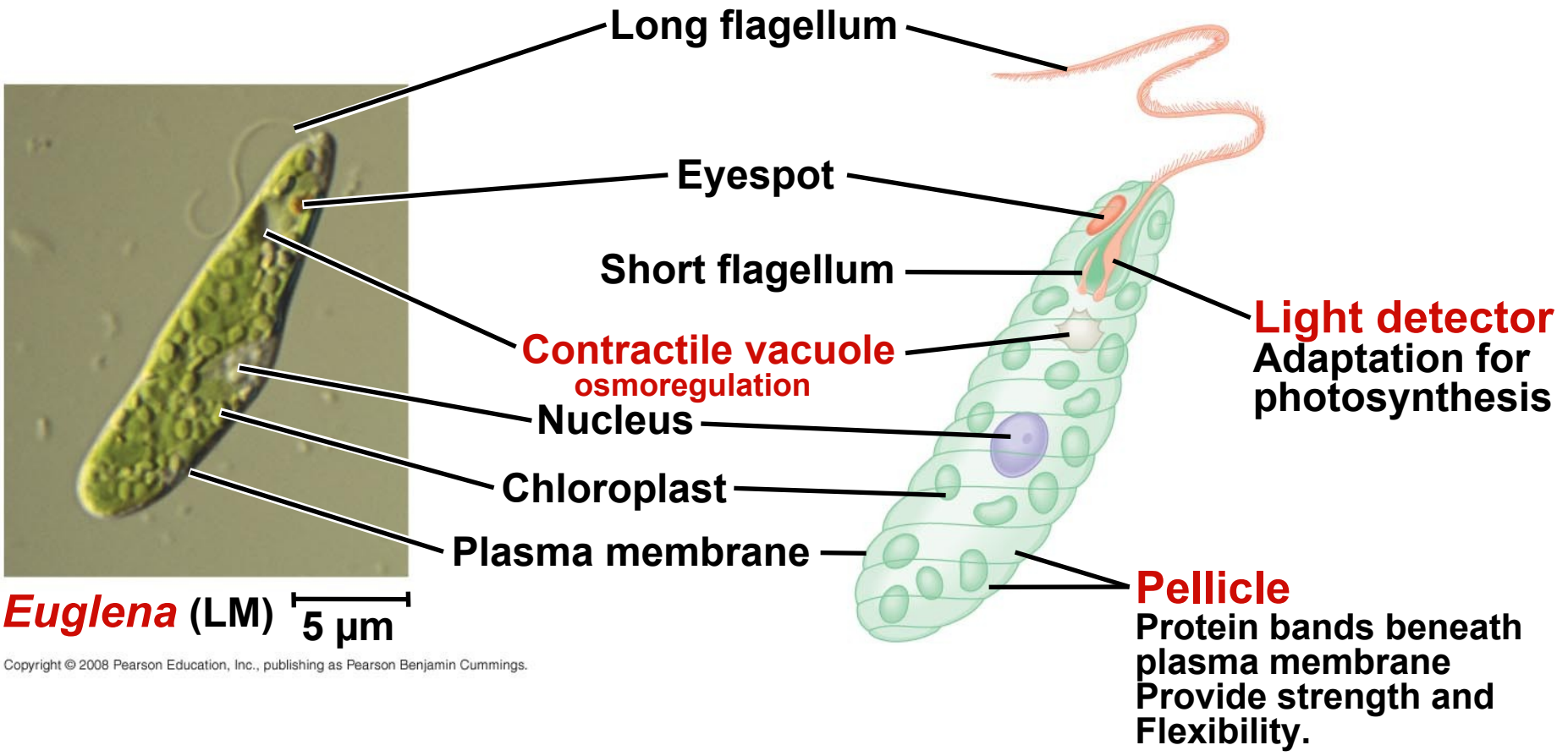


Trypanannnosoma - the Kinetoplastid that causes Sleeping Sickness

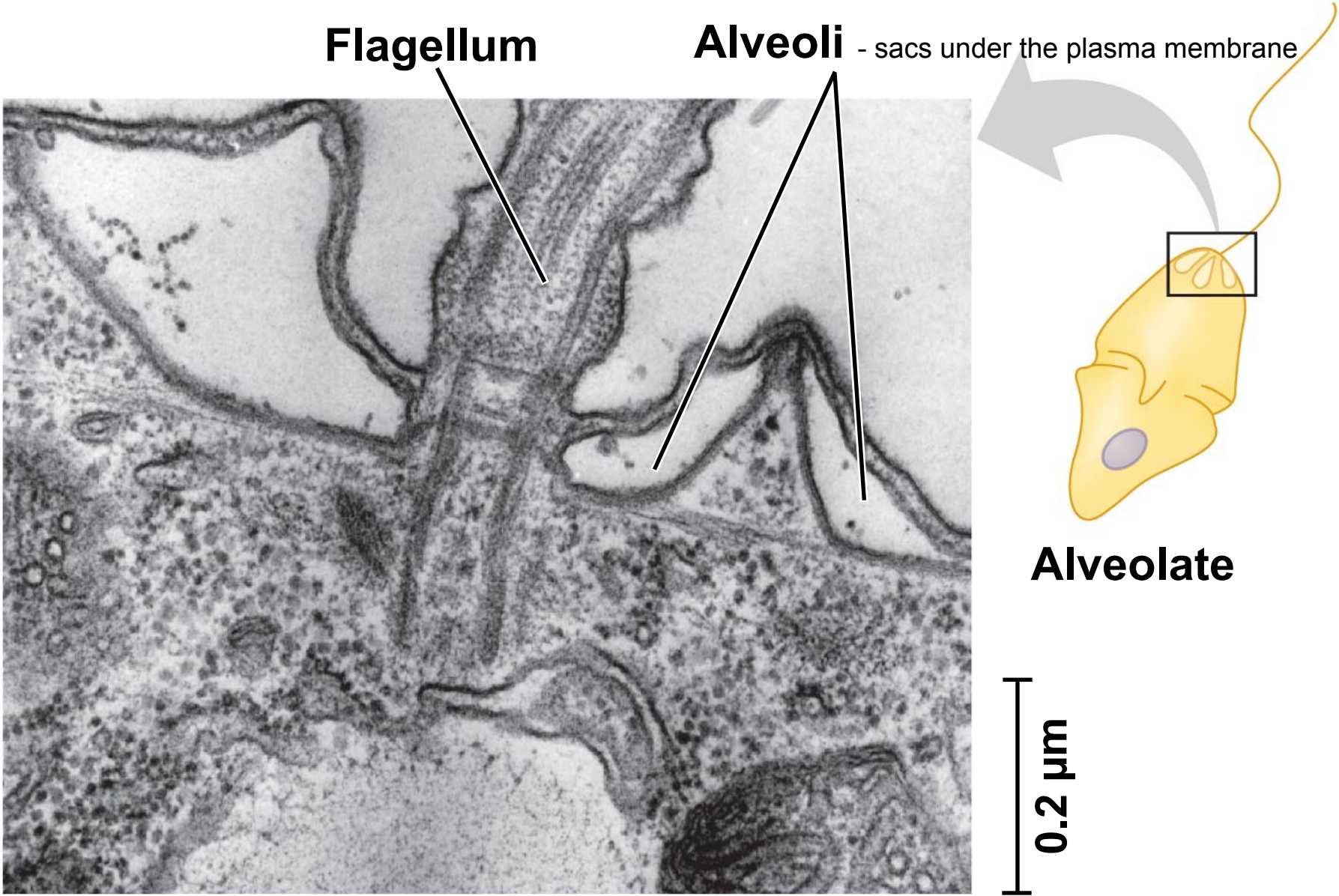


Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings.

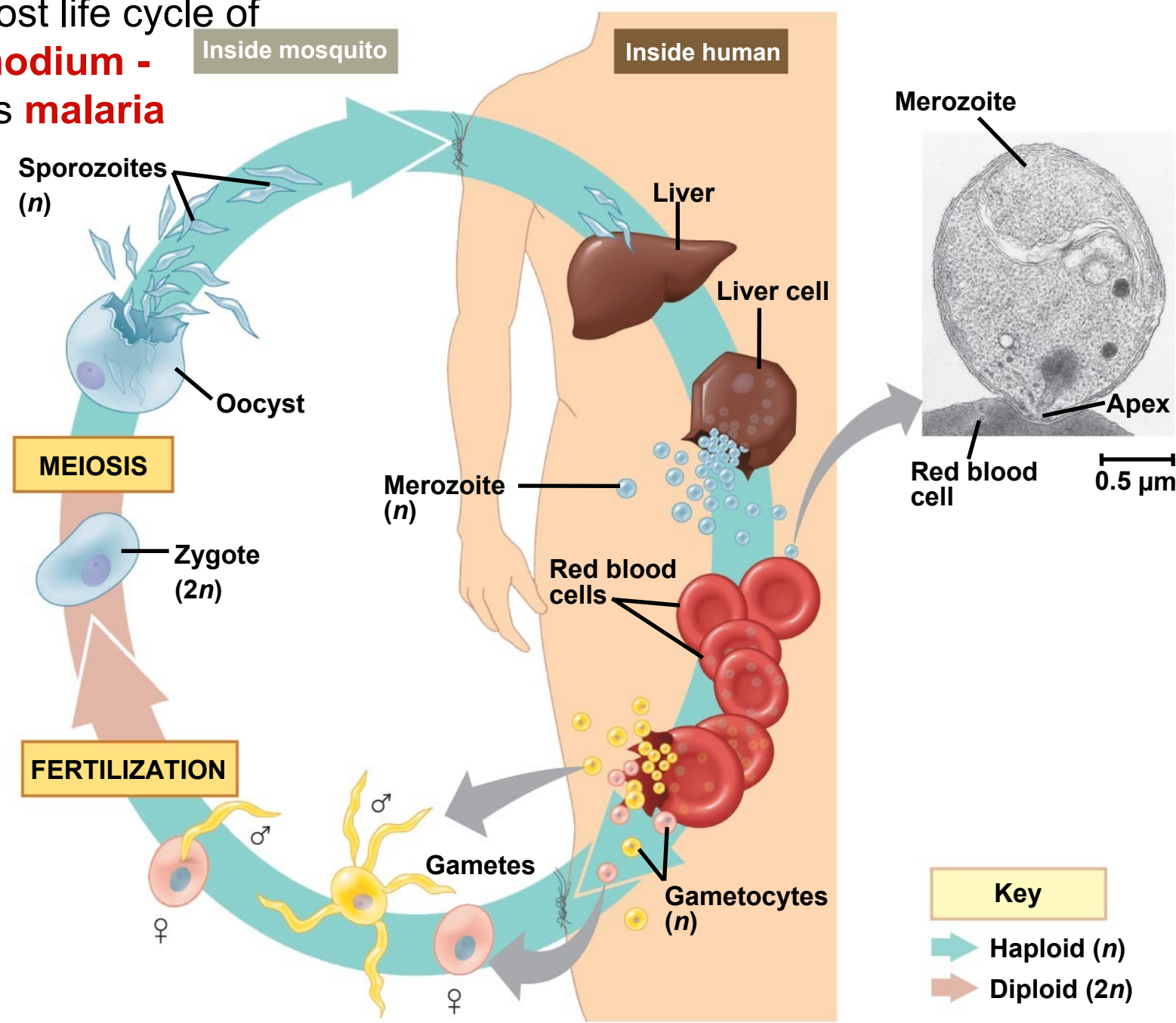
Euglenids can be **both autotrophic** and **heterotrophic**



Dinoflagellates some are **bioluminescent** /cause “**Red Tides**”



Two host life cycle of **Plasmodium** - causes **malaria**

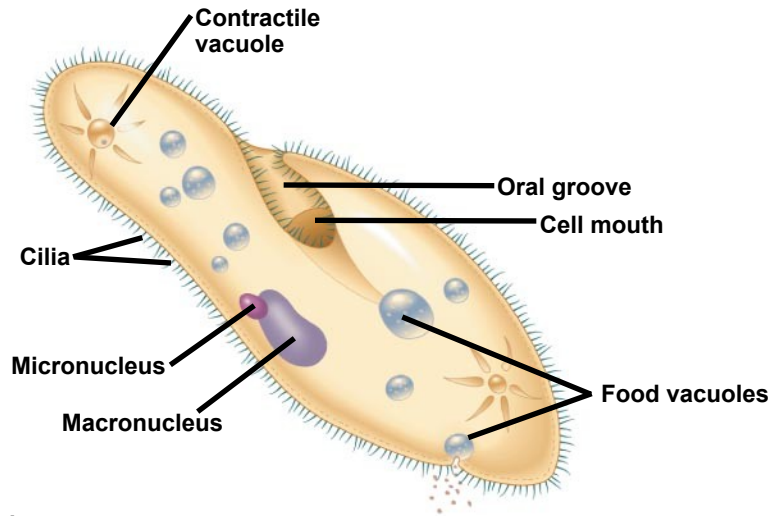


Ciliates: Paramecium

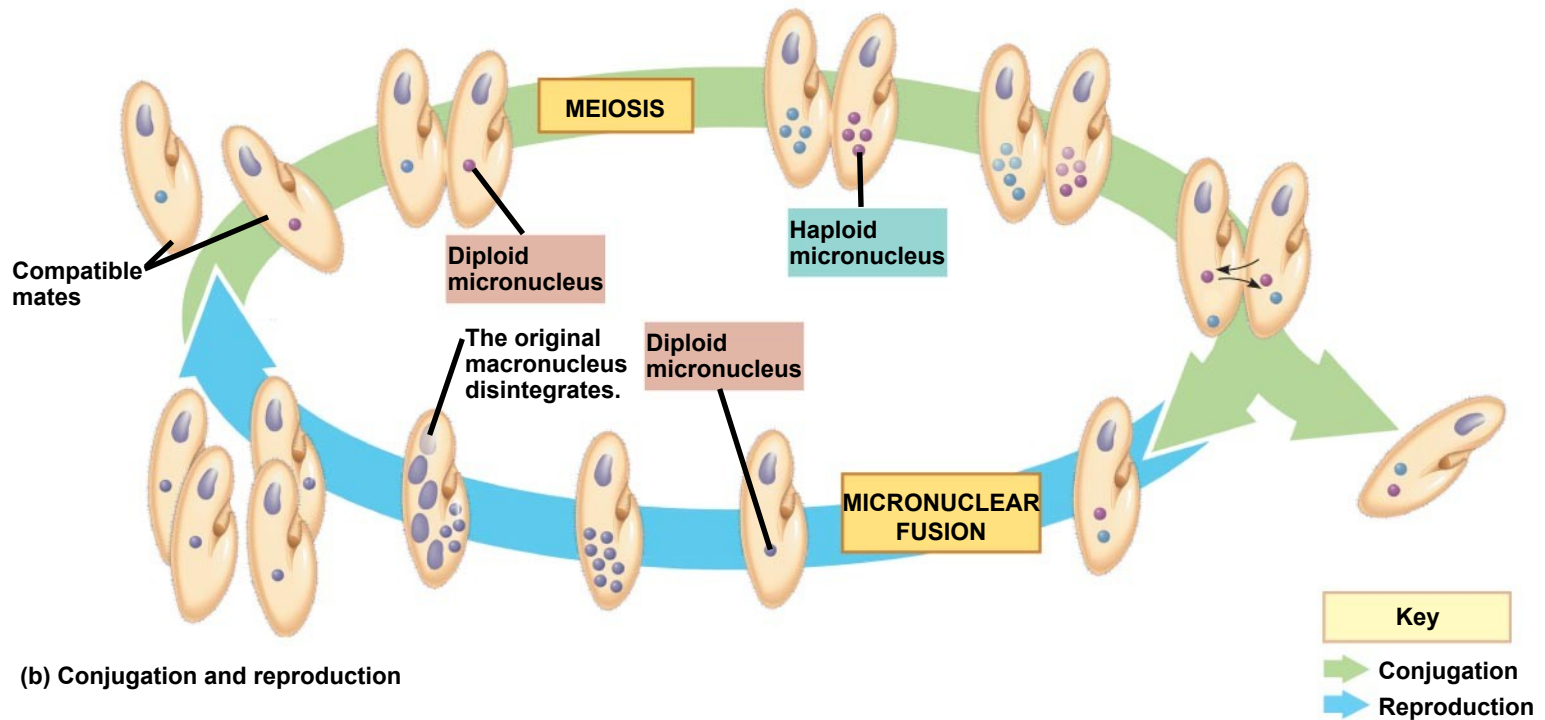
- ***Ciliates***, a large varied group of protists, are named for their use of *cilia to move and feed*.
- They have large *macronuclei* and small *micronuclei* - micronuclei function during *conjugation*, a sexual process that produces genetic variation.
- Conjugation is separate from reproduction, which generally occurs by binary fission.

Paramecium

50 μm



(a) Feeding, waste removal, and water balance

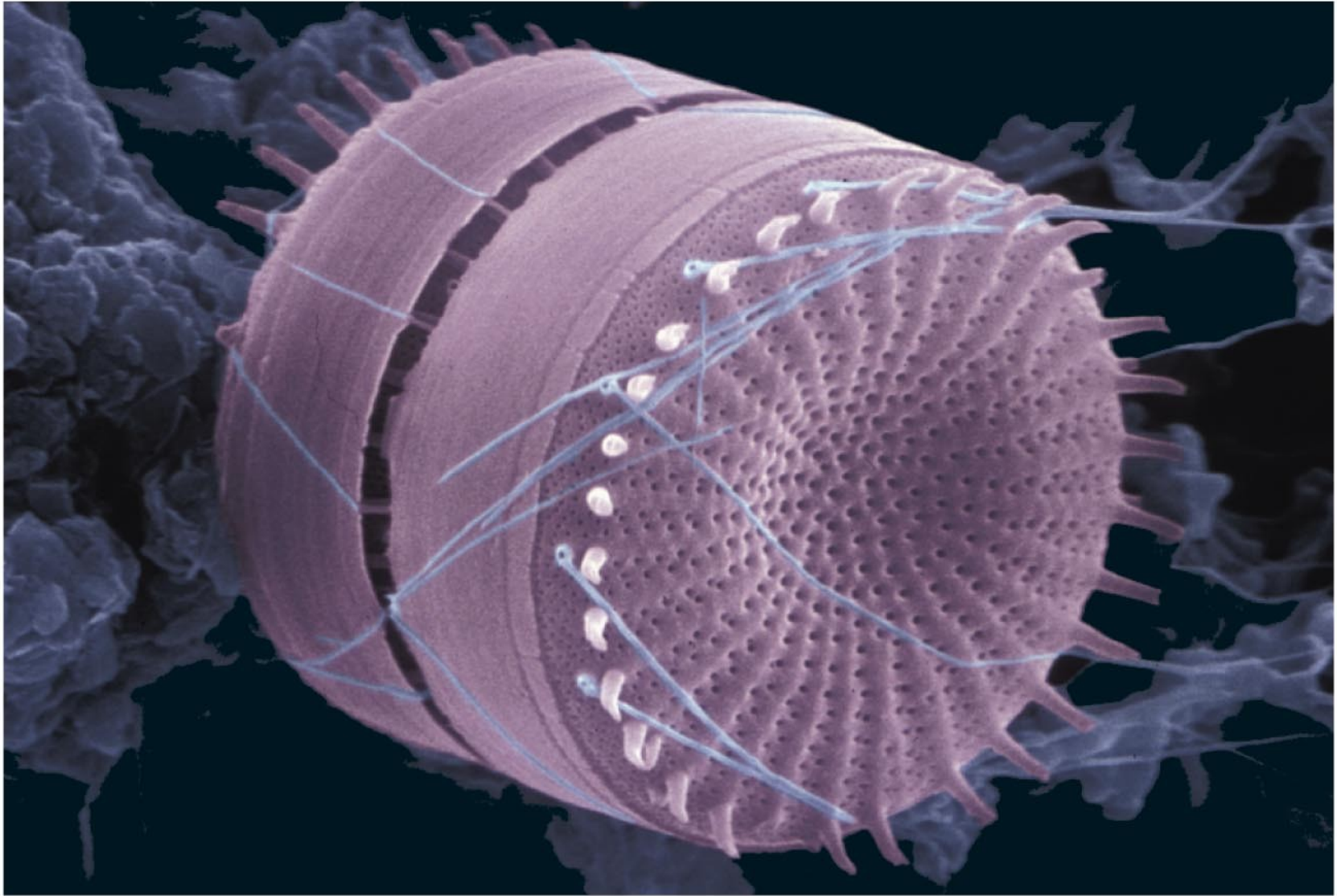


(b) Conjugation and reproduction

Diatoms

- *Diatoms* are *unicellular algae* with a unique two-part, *glass-like wall* of hydrated *silica*.
- Diatoms usually reproduce asexually, and occasionally sexually.
- Diatoms are *a major component of phytoplankton* and are highly diverse.
- Fossilized *diatom walls* compose much of the sediments known as *diatomaceous earth*.

Freshwater Diatom with glass-like wall of silica



3 μm

Golden Algae

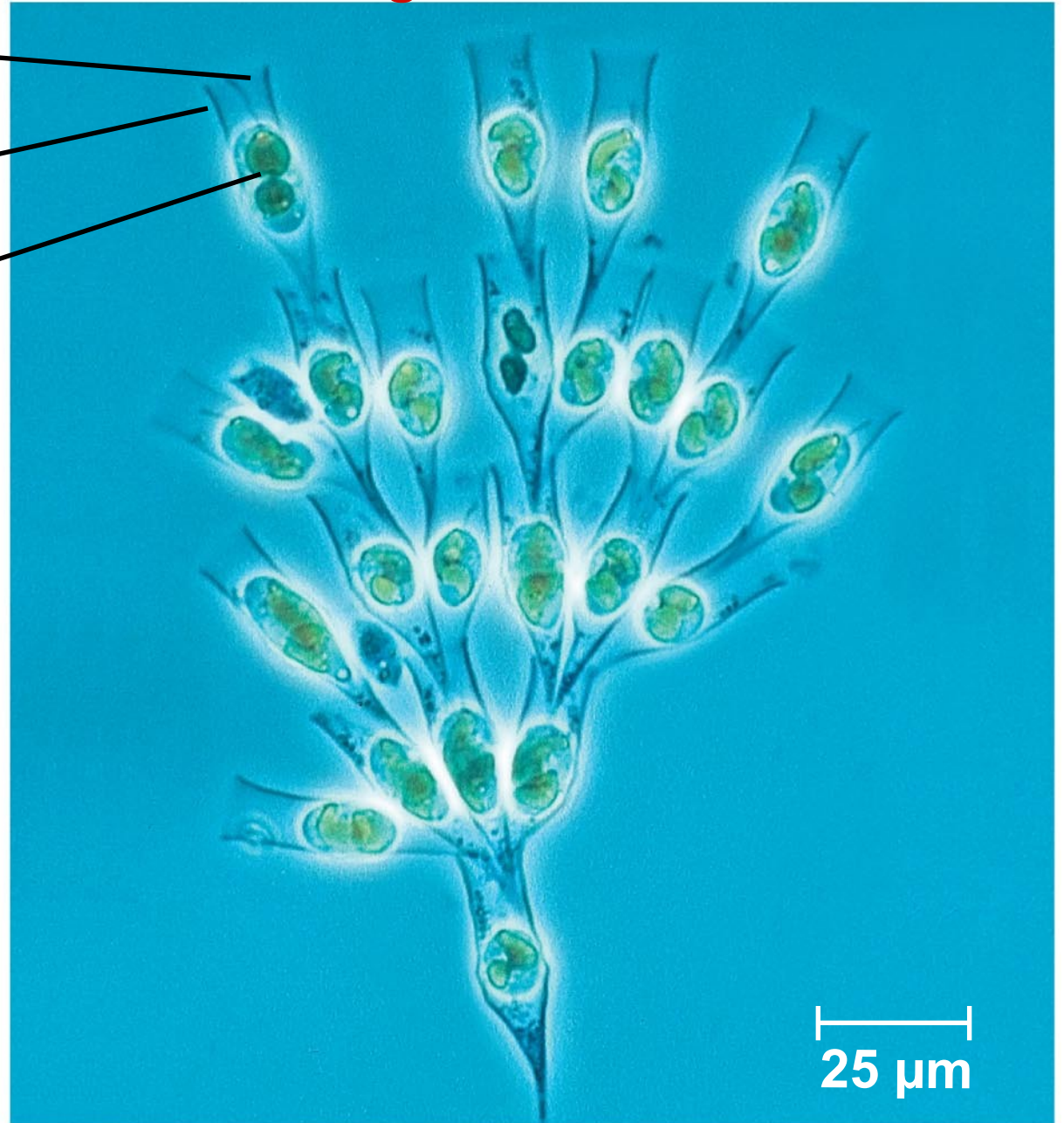
- **Golden algae** are named for their *color*, which results from yellow and brown *carotenoids*.
- The *cells* of golden algae are typically *biflagellated*, with both flagella near one end.
- All golden algae are *photosynthetic*, and some are also heterotrophic.
- Most are unicellular, but some are colonial.

Colonial Golden Algae

Flagellum

Outer container

Living cell



25 μm

Brown Algae

- **Brown algae** are the *largest* and most complex algae. All are *multicellular*, and most are marine.
- Brown algae include many species commonly called “*seaweeds*.” Giant seaweeds called *kelps* live in deep parts of the ocean.
- Brown algae have the *most complex* multicellular anatomy of all algae.
- The *algal body* is plantlike but lacks true roots, stems, and leaves and is called a *thallus*.
- The *rootlike holdfast* anchors the *stemlike stipe*, which in turn supports the *leaflike blades*.

Brown Algae

Blade

Stipe

Holdfast



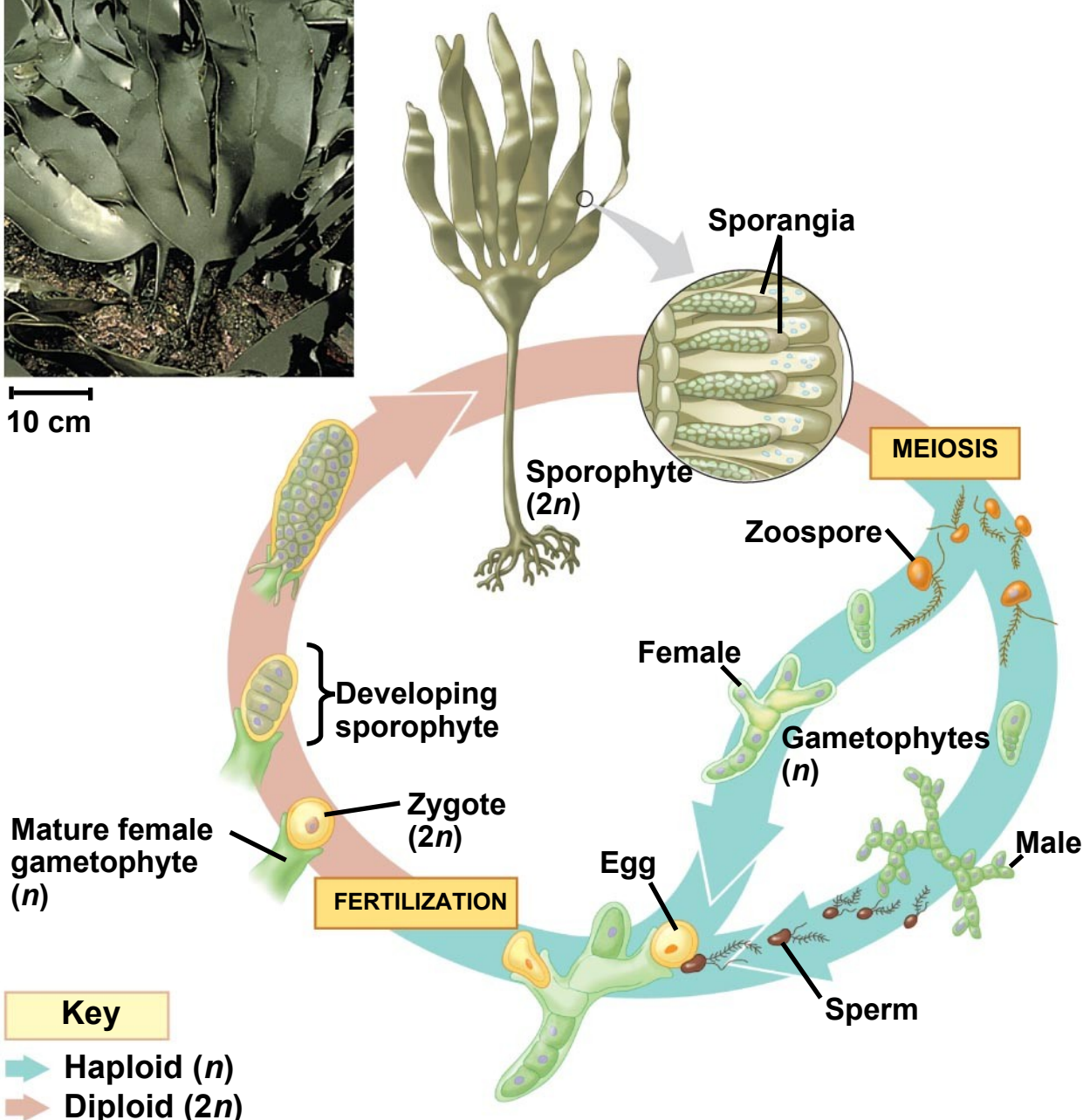
Alternation of Generations

- A variety of life cycles have evolved among the multicellular algae.
- The most complex life cycles include an ***alternation of generations***, the alternation of multicellular haploid (*gametophyte*) and diploid forms (*sporophyte*).
- ***Heteromorphic*** generations are structurally *different*.
- ***Isomorphic*** generations look *similar*.

Brown Algae



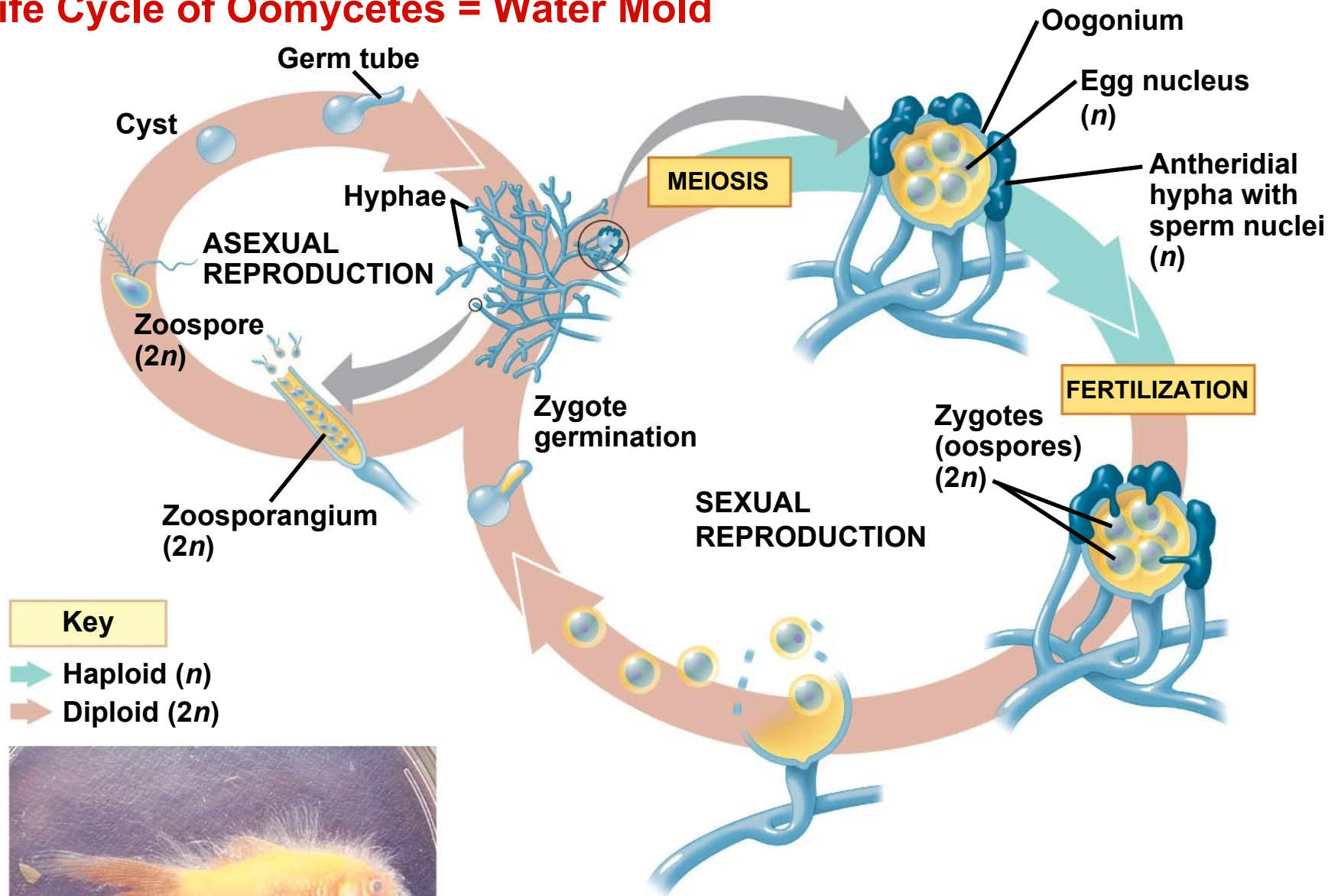
Alternation of Generations



Oomycetes (Water Molds and Their Relatives)

- ***Oomycetes*** include water molds, white rusts, and downy mildews... *Fungus-like*.
- They were once considered fungi based on morphological studies.
- Most oomycetes are *decomposers* or *parasites*
- They have filaments (*hyphae*) that facilitate *nutrient uptake*.
- Their ecological impact can be great, as in *Phytophthora infestans* causing potato blight.

Life Cycle of Oomycetes = Water Mold



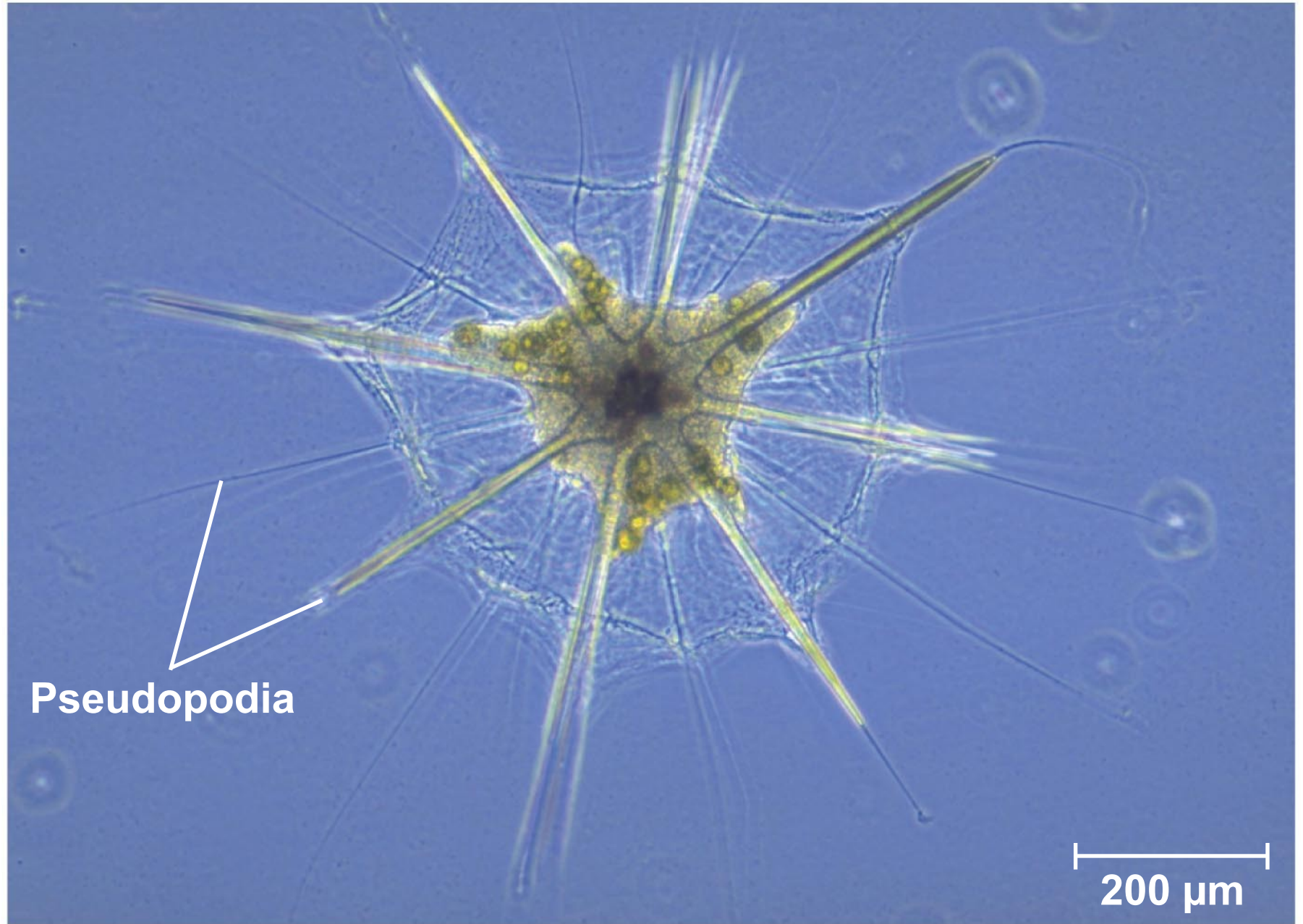
Concept 28.4: Rhizarians are a diverse group of protists defined by DNA similarities

- DNA evidence supports *Rhizaria* as a monophyletic clade.
- *Amoebas* move and feed by *pseudopodia*; some but not all belong to the clade Rhizaria.
- Rhizarians include *forams* and *radiolarians*.

Protists: Forams and Radiolarians

- ***Foraminiferans***, or ***forams***, are named for porous, generally multichambered ***shells***, called ***tests***. ***Pseudopodia*** extend through the pores in the test.
- *Foram tests in marine sediments form an extensive fossil record.*
- ***Radiolarians*** have tests fused into one delicate piece, made of silica.
- Radiolarians use their ***pseudopodia*** to ***engulf*** microorganisms by ***phagocytosis***.

Radiolarian Protist



Concept 28.5: Red algae and green algae are the closest relatives of land plants

- Over a billion years ago, a heterotrophic protist acquired a cyanobacterial endosymbiont.
- The photosynthetic descendants of this ancient protist evolved into red algae and green algae.
- *Land plants are descended from the green algae.*
- ***Archaeplastida*** is a supergroup used by some scientists and includes *red algae*, *green algae*, and *land plants*.

Red Algae

- **Red algae** are reddish in color due to an *accessory pigment* call *phycoerythrin*, which masks the green of chlorophyll.
- The color varies from greenish-red in shallow water to dark red or almost black in deep water. *Red algae are the most abundant large algae in coastal waters of the tropics.*
- Red algae are usually *multicellular*; the largest are *seaweeds*.

Red Algae : This species has a delicate filamentous form

▶ ***Bonnemaisonia
hamifera***



8 mm

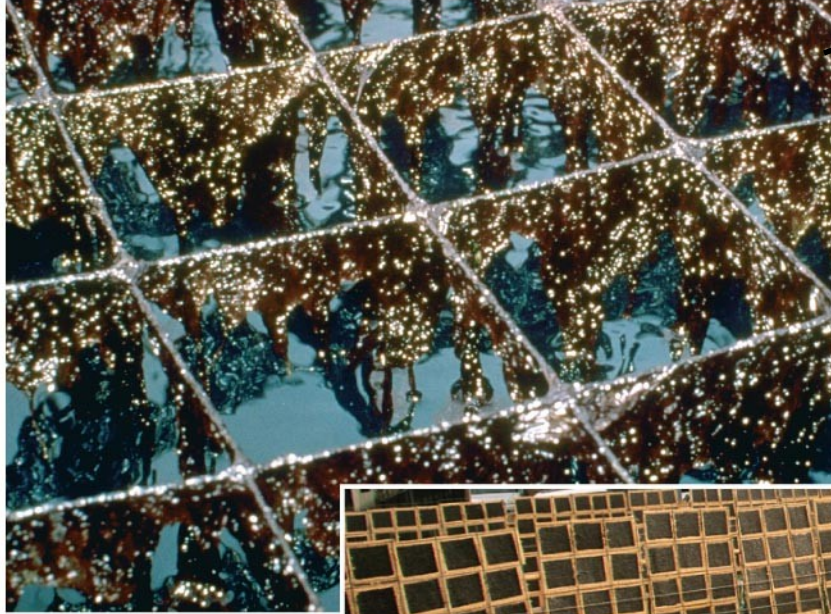
Red Algae: This leafy species is edible

20 cm

A horizontal scale bar with vertical end caps, indicating a length of 20 cm.

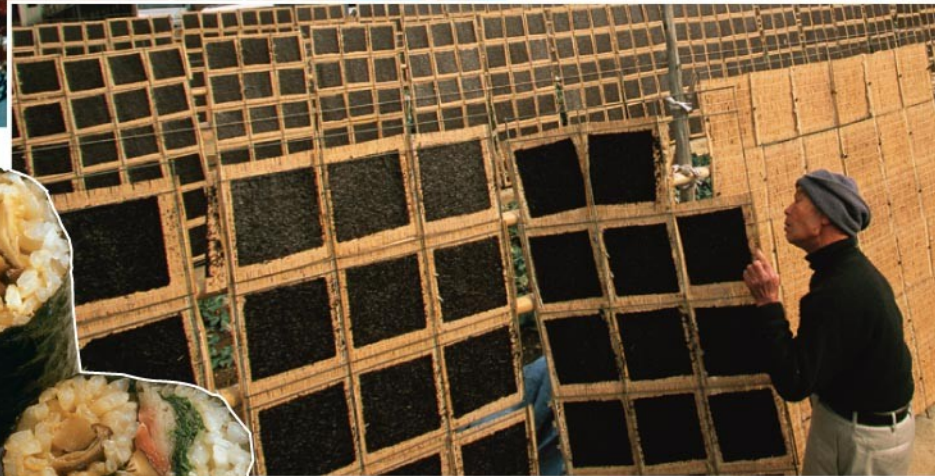
◀ **Dulse (*Palmaria palmata*)**

▼ Nori. The **red alga** *Porphyra* is the source of a traditional **Japanese food**.



The seaweed is grown on nets in shallow coastal waters.

The harvested seaweed is spread on bamboo screens to dry.



Paper-thin, glossy sheets of nori make a mineral-rich wrap for rice, seafood, and vegetables in sushi.

Green Algae

- **Green algae** are named for their grass-green chloroplasts.
- *Plants are descended from the green algae.*
- The two main groups are *chlorophytes* and *charophyceans*.
- Most chlorophytes live in fresh water, although many are marine. Other chlorophytes live in damp soil, as *symbionts in lichens*, or in snow.
- Chlorophytes include unicellular, colonial, and multicellular forms.

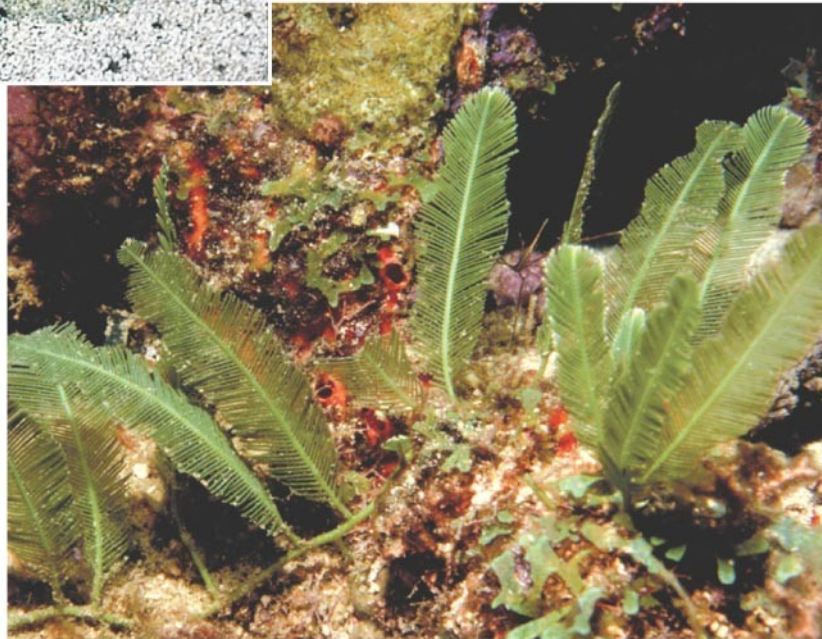
Green Algae



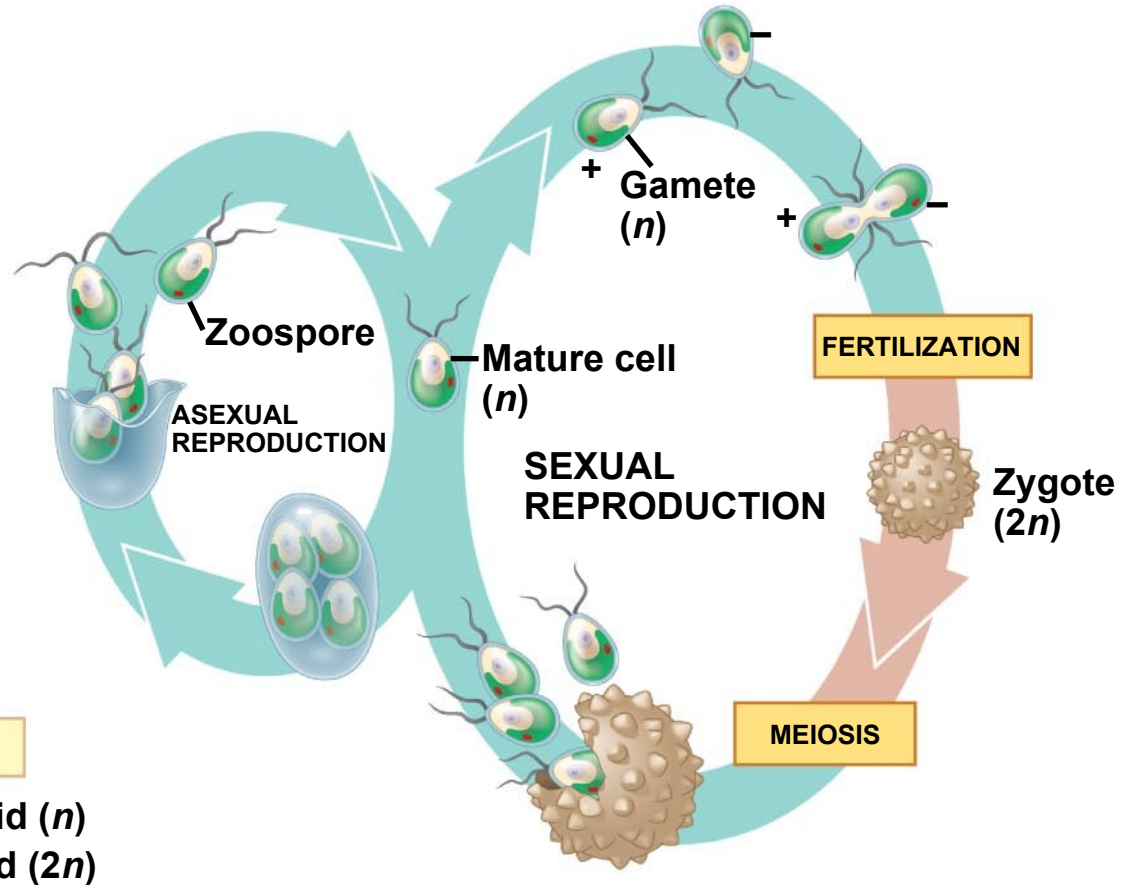
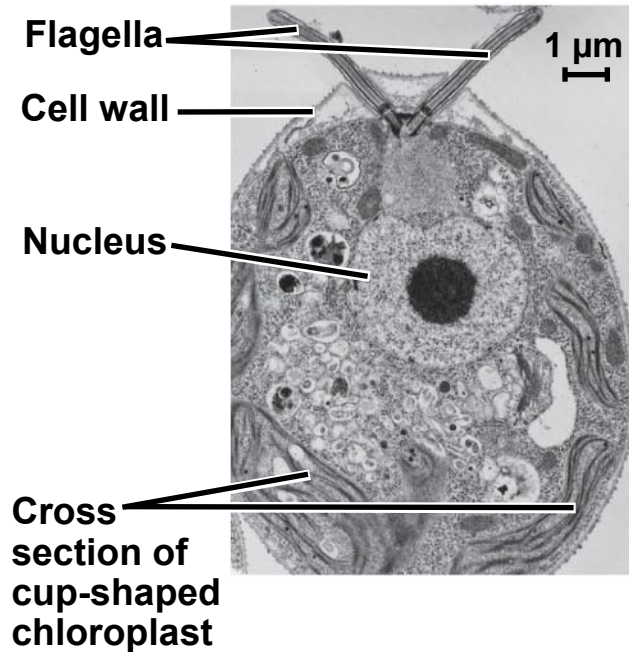
2 cm

(a) *Ulva*, or sea lettuce
edible seaweed

(b) *Caulerpa*, an
intertidal chloro-
phyte



Most chlorophytes have complex life cycles with both sexual and asexual reproductive stages:

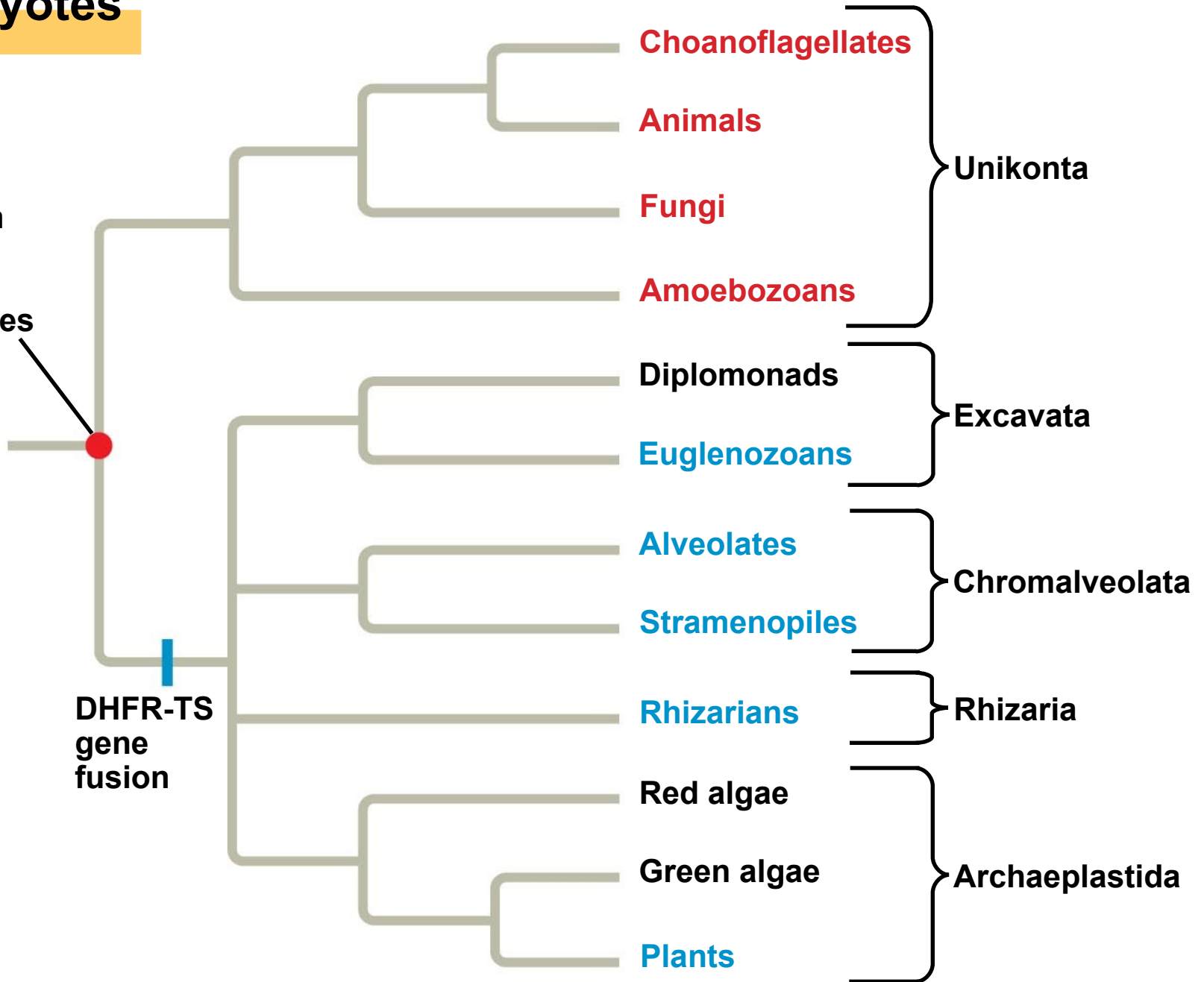


Concept 28.6: Unikonts include protists that are closely related to fungi and animals

- The supergroup **Unikonta** includes animals, fungi, and some protists.
- This group includes two clades: the amoebozoans and the opisthokonts (animals, fungi, and related protists)
- The root of the eukaryotic tree remains controversial.

Eukaryotes

Common
ancestor
of all
eukaryotes



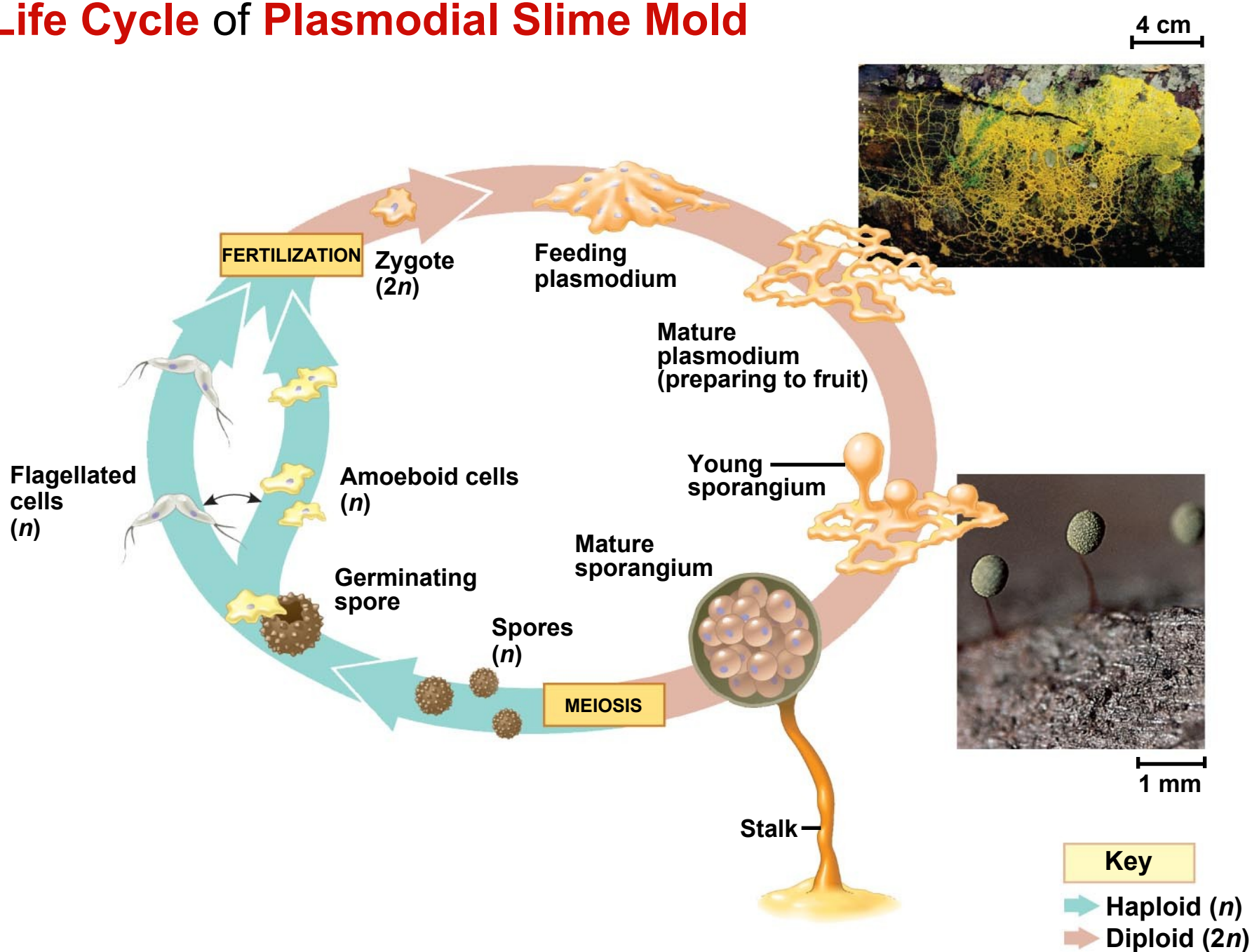
Amoebozoans

- ***Amoebozoans*** are *amoeba* that have lobe- or tube-shaped, rather than threadlike, *pseudopodia*.
- They include gymnamoebas, entamoebas, and slime molds.

Slime Molds > fungus-like

- Slime molds, or mycetozoans, were once thought to be fungi. Molecular systematics places slime molds in the clade Amoebozoa.
- Many species of **plasmodial slime molds** are brightly pigmented, usually yellow or orange.
- At one point in the life cycle, plasmodial slime molds form a mass called a **plasmodium** (not to be confused with malarial *Plasmodium*).
- The plasmodium is undivided by membranes and contains many diploid nuclei. It **extends pseudopodia through decomposing material, engulfing food by phagocytosis**.

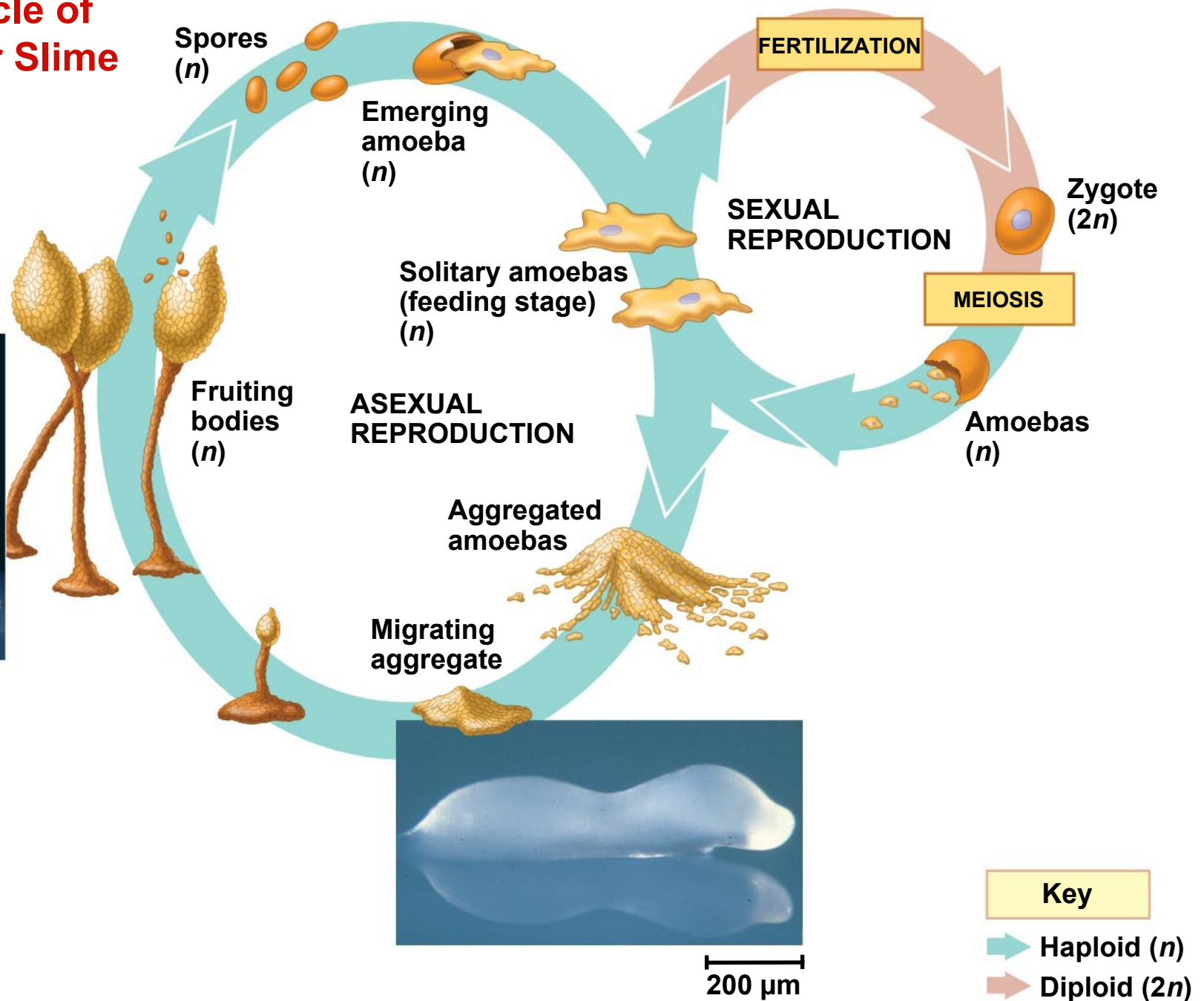
Life Cycle of Plasmodial Slime Mold



-
- ***Cellular slime molds*** form multicellular aggregates in which cells are separated by their membranes.
 - ***Cells*** feed individually, but ***can aggregate*** to form a fruiting body.
 - *Dictyostelium discoideum* is an experimental model for studying the ***evolution of multicellularity***.

Life Cycle of Cellular Slime Mold

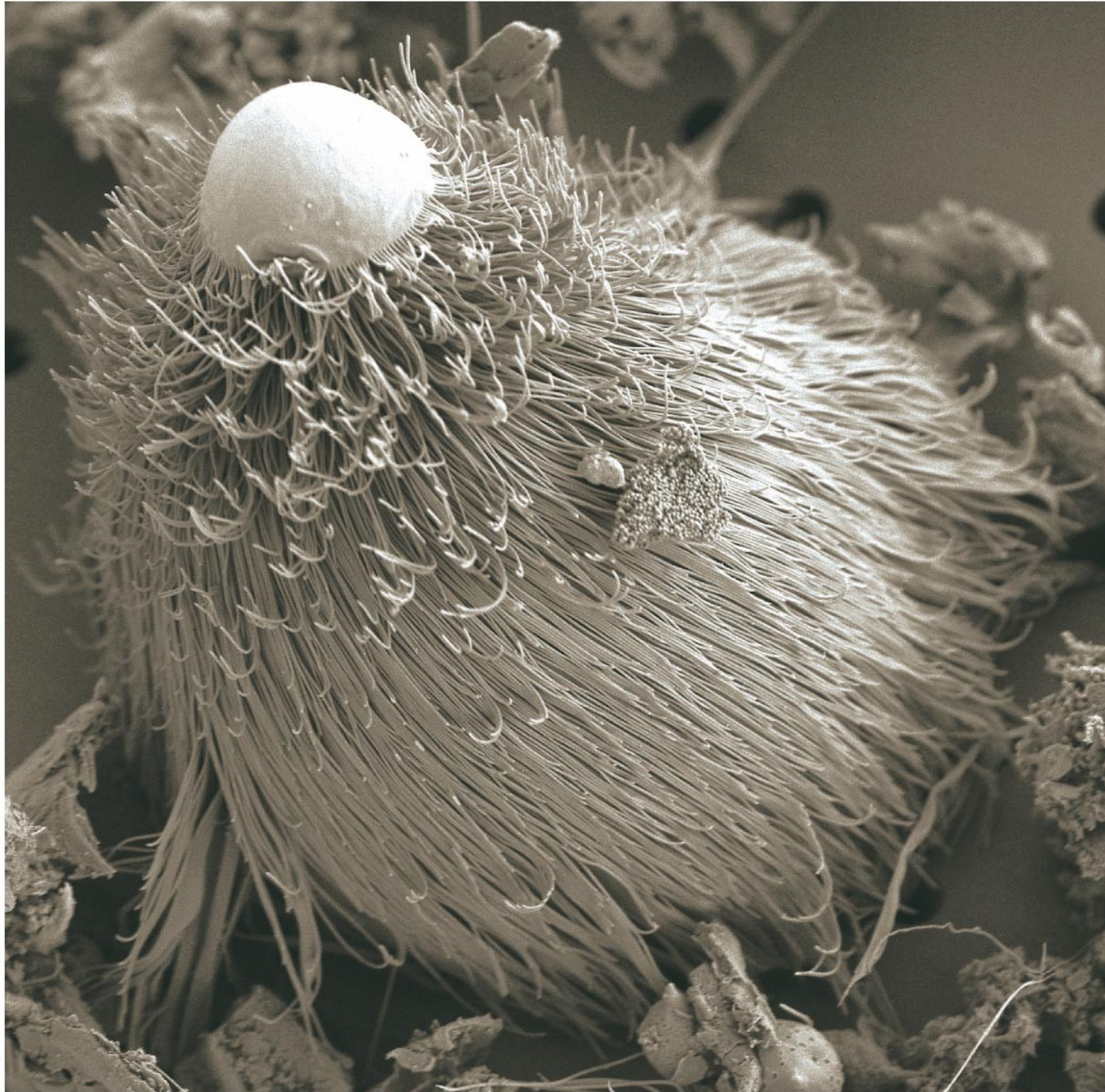
600 μm



Protists play key roles in ecological relationships

- Protists are found in diverse *aquatic* environments.
- Protists often play the role of *symbiont* or *producer*.
- Some ***protist symbionts ++*** benefit their hosts
 - *Dinoflagellates* nourish coral polyps that build reefs.
 - *Hypermastigotes* digest cellulose in the gut of termites.

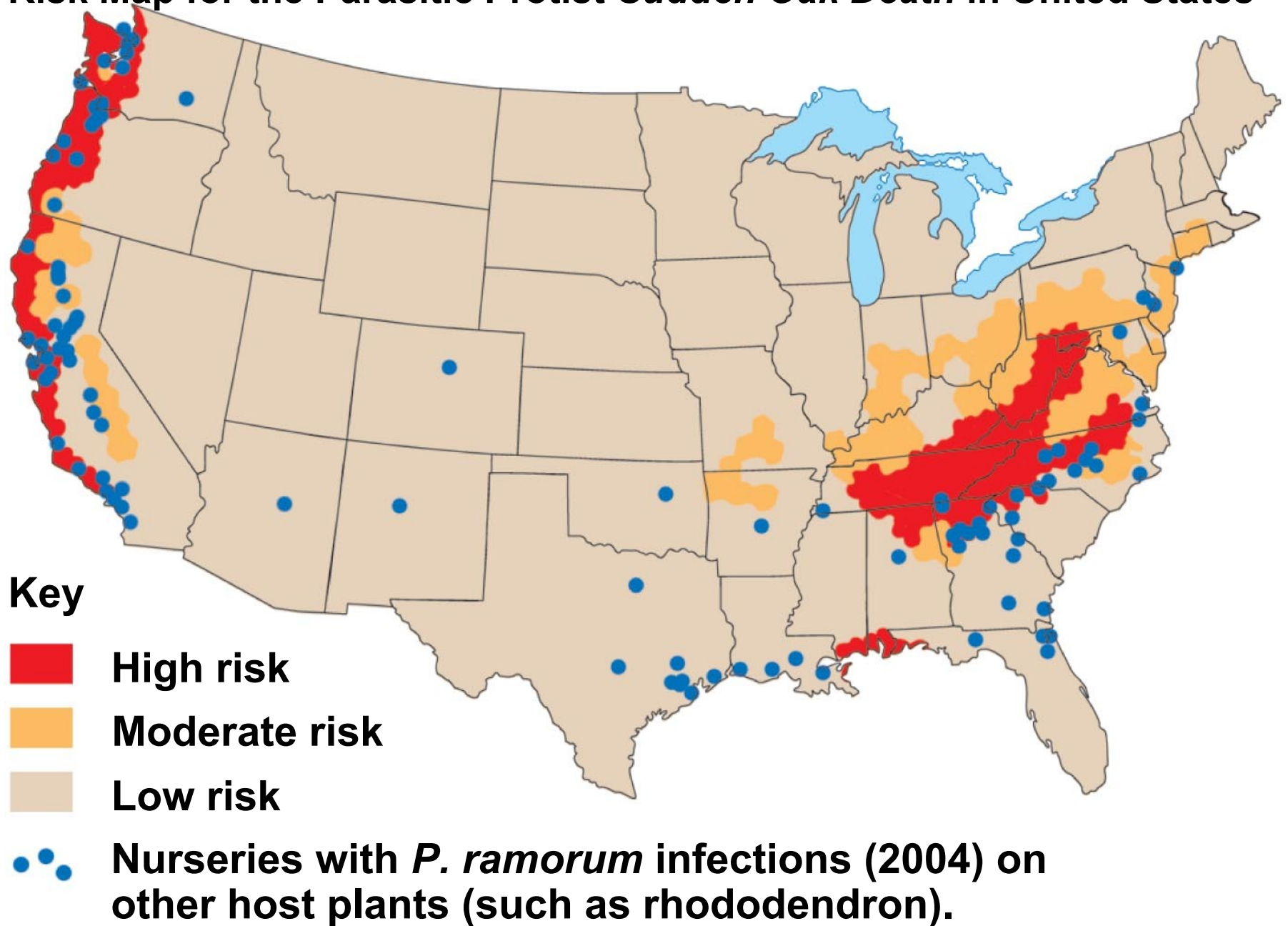
Protist Symbiont ++



10 μm

-
- Some *protists* are *parasitic + - symbionts*
 - *Plasmodium* causes *malaria*.
 - *Pfesteria shumwayae* is a *dinoflagellate* that causes fish kills... *toxic red tides* ..
 - *Phytophthora ramorum* causes sudden oak death.

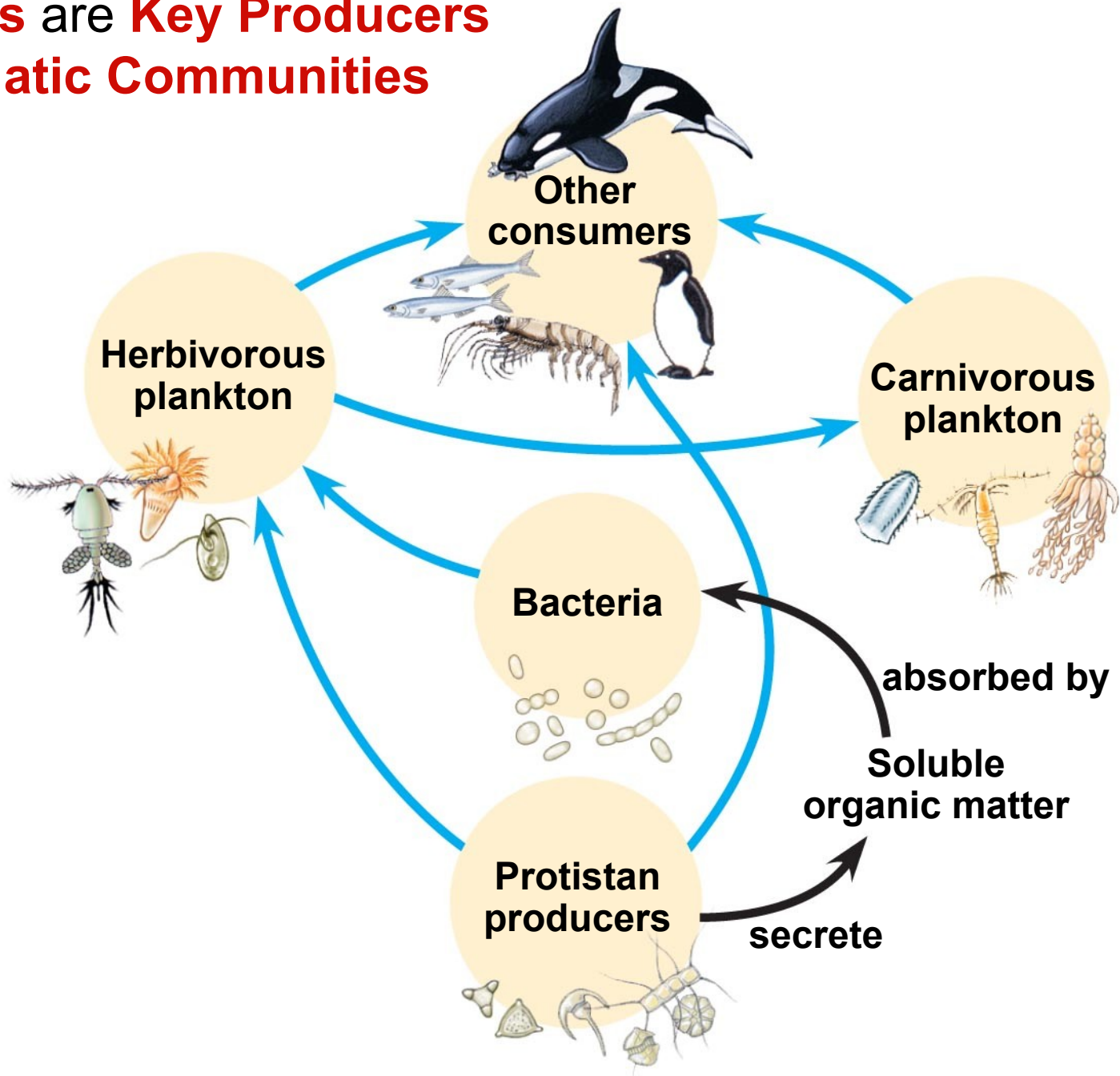
Risk Map for the Parasitic Protist *Sudden Oak Death* in United States



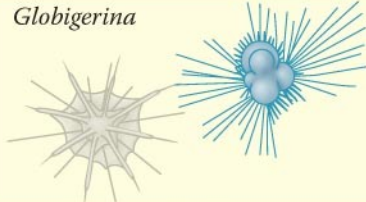
Photosynthetic Protists

- Many protists are important *producers* that obtain energy from the sun.
- In *aquatic environments*, photosynthetic protists and prokaryotes are the main producers.
- The availability of nutrients can affect the concentration of protists.

Protists are Key Producers in Aquatic Communities



Review

Key Concept/Eukaryote Supergroup	Major Clades	Key Morphological Characteristics	Specific Examples
Concept 28.2 Excavates include protists with modified mitochondria and protists with unique flagella (pp. 580–581)	Diplomonads and parabasalids Euglenozoans Kinetoplastids Euglenids	Modified mitochondria Spiral or crystalline rod inside flagella	<i>Giardia</i> , <i>Trichomonas</i>  <i>Trypanosoma</i> , <i>Euglena</i> 
Concept 28.3 Chromalveolates may have originated by secondary endosymbiosis (pp. 582–589)	Alveolates Dinoflagellates Apicomplexans Ciliates Stramenopiles Oomycetes Diatoms Golden algae Brown algae	Membrane-bounded sacs (alveoli) beneath plasma membrane Hairy and smooth flagella	<i>Pfiesteria</i> , <i>Plasmodium</i> , <i>Paramecium</i>  <i>Phytophthora</i> , <i>Laminaria</i> 
Concept 28.4 Rhizarians are a diverse group of protists defined by DNA similarities (pp. 589–590)	Forams Radiolarians	Amoebas with threadlike pseudopodia and a porous shell Amoebas with threadlike pseudopodia radiating from central body	<i>Globigerina</i> 
Concept 28.5 Red algae and green algae are the closest relatives of land plants (pp. 590–592—Archaeplastida)	Red algae Green algae Land plants	Phycoerythrin (accessory pigment) Plant-type chloroplasts (See Chapters 29 and 30.)	<i>Porphyra</i>  <i>Chlamydomonas</i> , <i>Ulva</i>  Mosses, ferns, conifers, flowering plants 
Concept 28.6 Unikonts include protists that are closely related to fungi and animals (pp. 593–596)	Amoebozoans Slime molds Gymnamoebas Entamoebas Opisthokonts	Amoebas with lobe-shaped pseudopodia (Highly variable; see Chapters 31–34.)	<i>Amoeba</i> , <i>Entamoeba</i> , <i>Dictyostelium</i>  Nucleariids, choanoflagellates, animals, fungi 

You should now be able to:

1. Explain why the kingdom Protista is no longer considered a legitimate taxon.
2. Explain the process of endosymbiosis and state what living organisms are likely relatives of mitochondria and plastids.
3. Distinguish between endosymbiosis and secondary endosymbiosis.
4. Name the five supergroups, list their key characteristics, and describe some representative taxa.